



CMPT 413 / 713: Natural Language Processing

Recurrent Neural Networks

LSTM and GRUs

How to model sequences using neural networks?

Fall 2021

2021-10-19

Adapted from slides from Danqi Chen, Karthik Narasimhan, and Justin Johnson

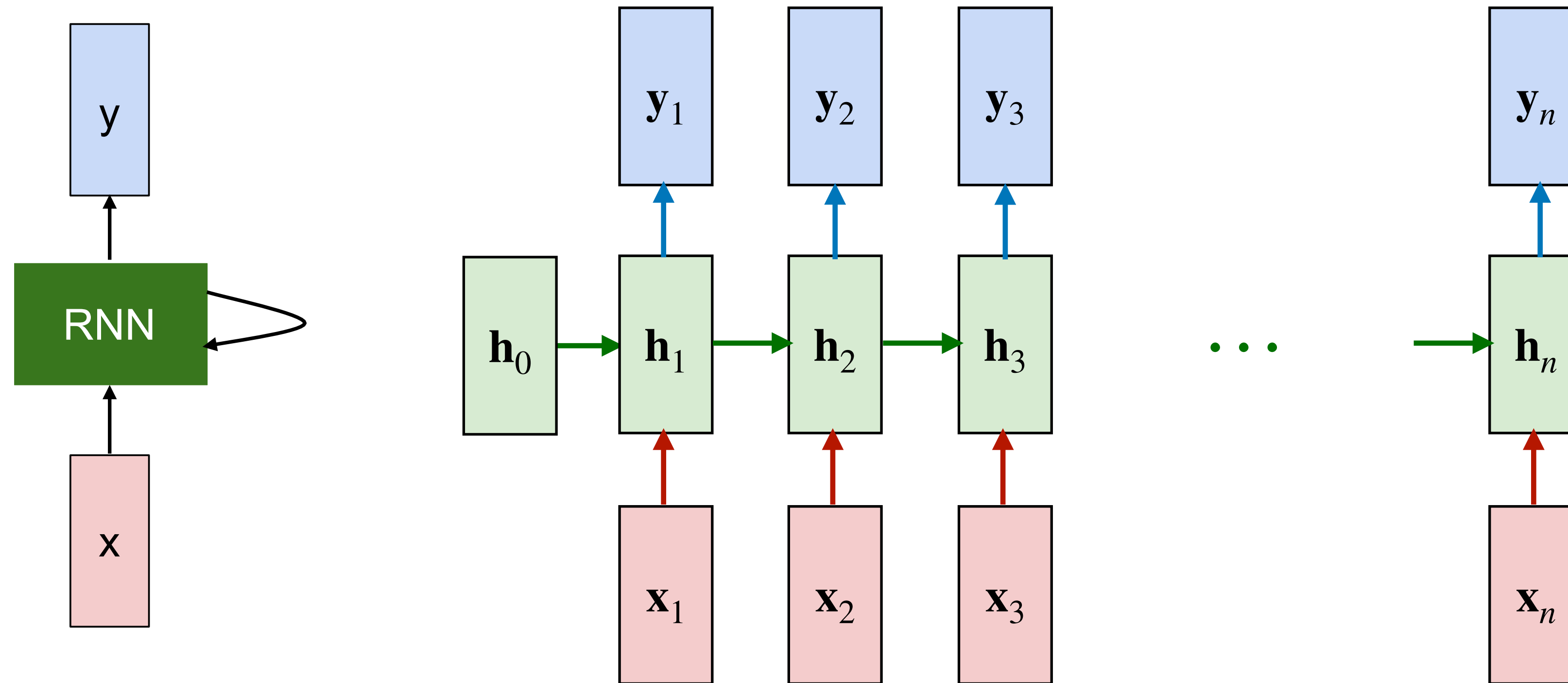
(Some slides adapted from Chris Manning, Abigail See, Andrej Karpathy)

Overview

- Review of Vanilla RNN
- Issues with Gradient Flows
- LSTMs and GRUs
- Applications
- Variants: Stacked RNNs, Bidirectional RNNs

Simple RNNs

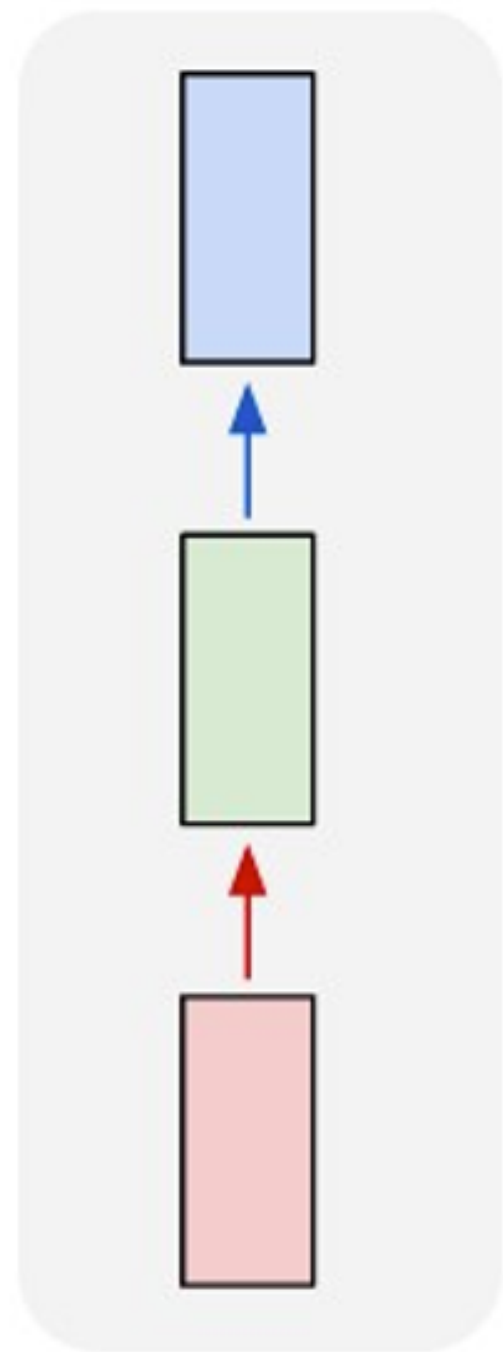
Recurrent Neural Networks (RNNs)



Structure of cell and weights are shared across time steps

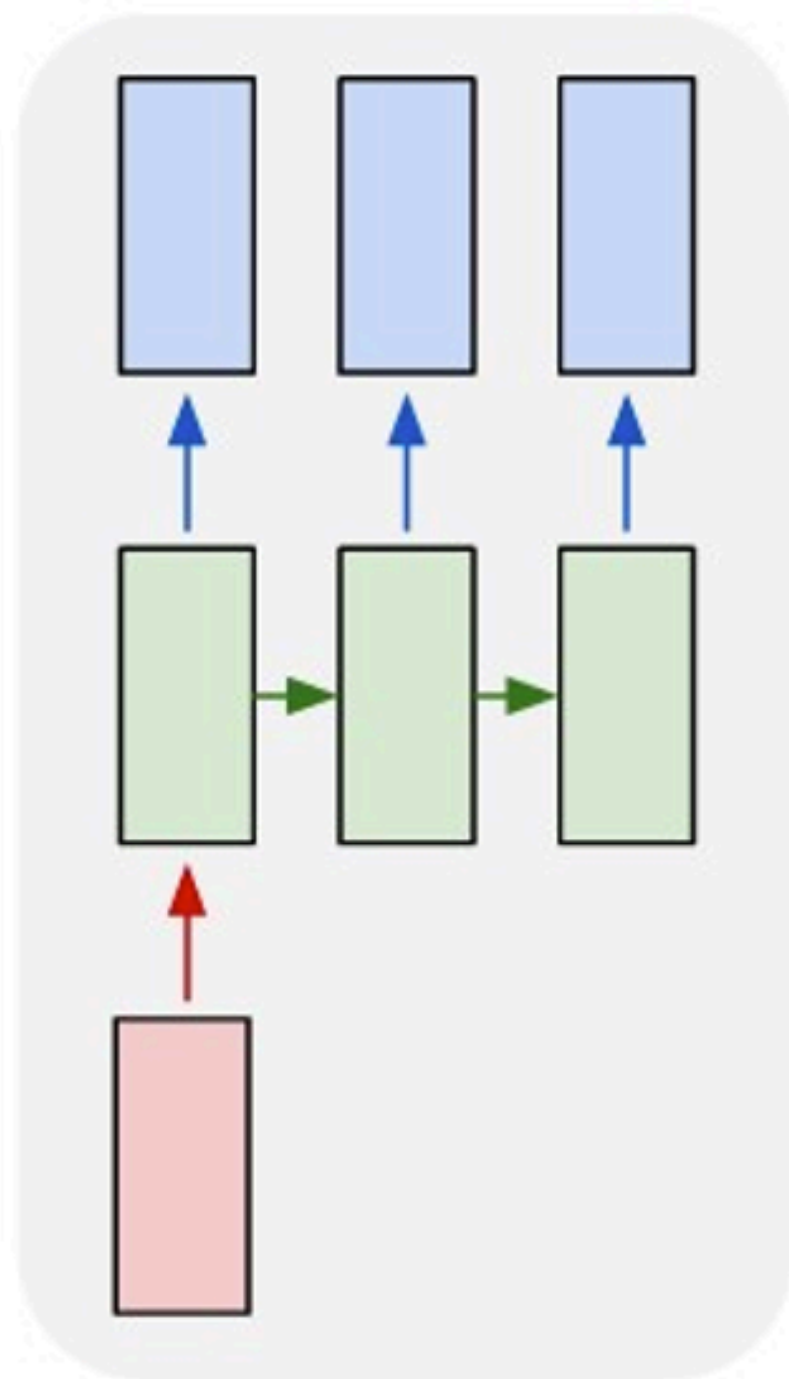
Types of sequence processing problems

one to one



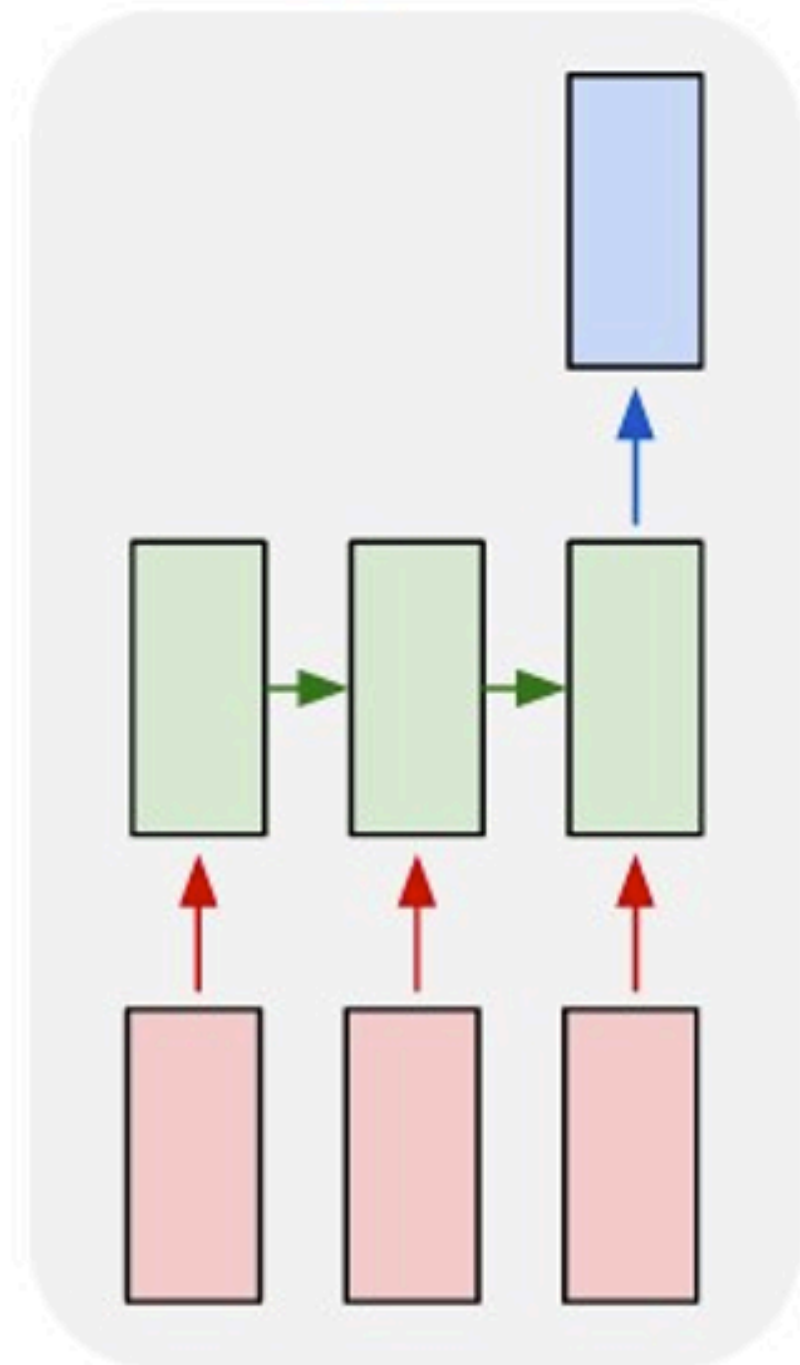
Vanilla
Neural
Networks

one to many



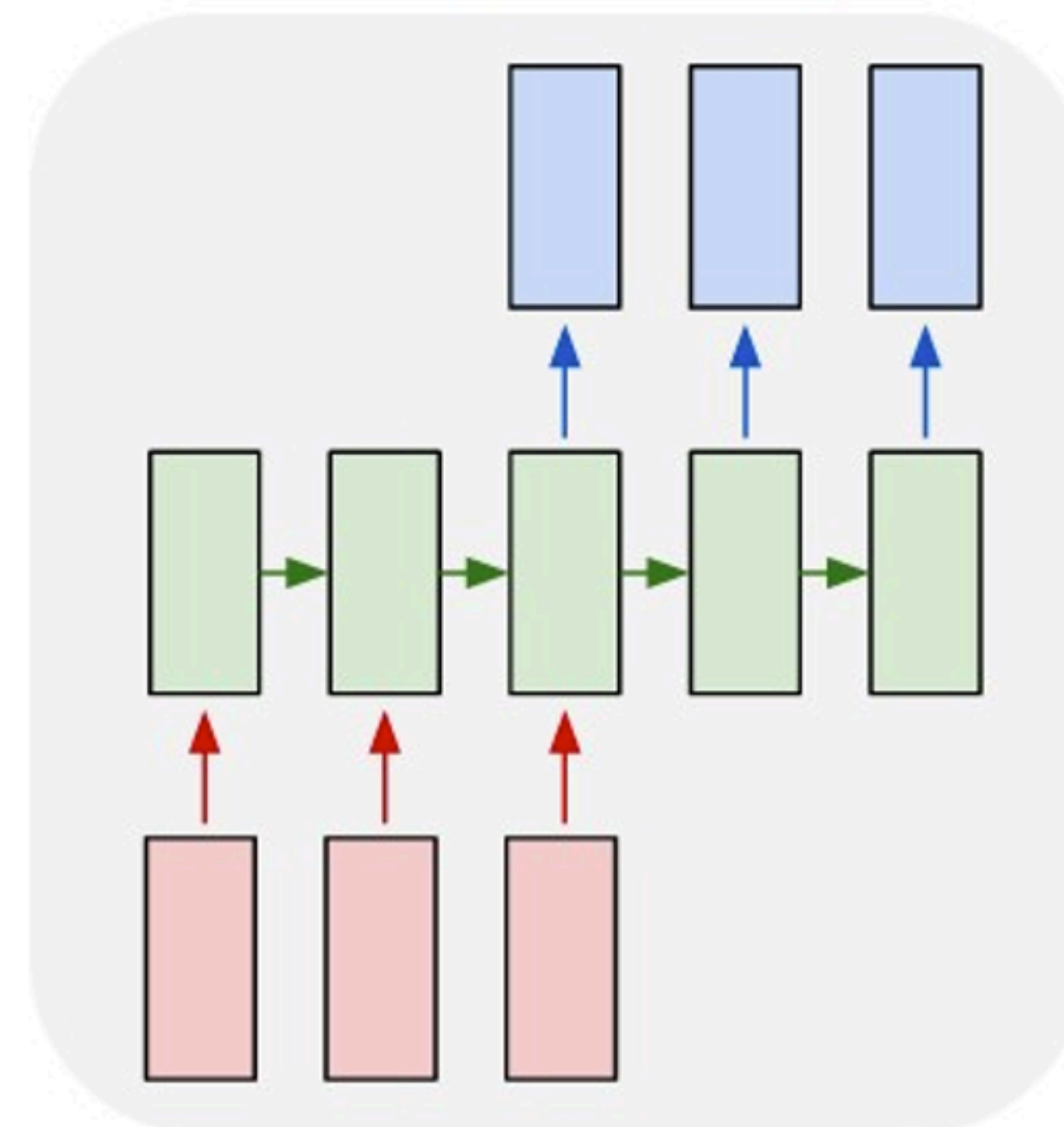
Text Generation

many to one



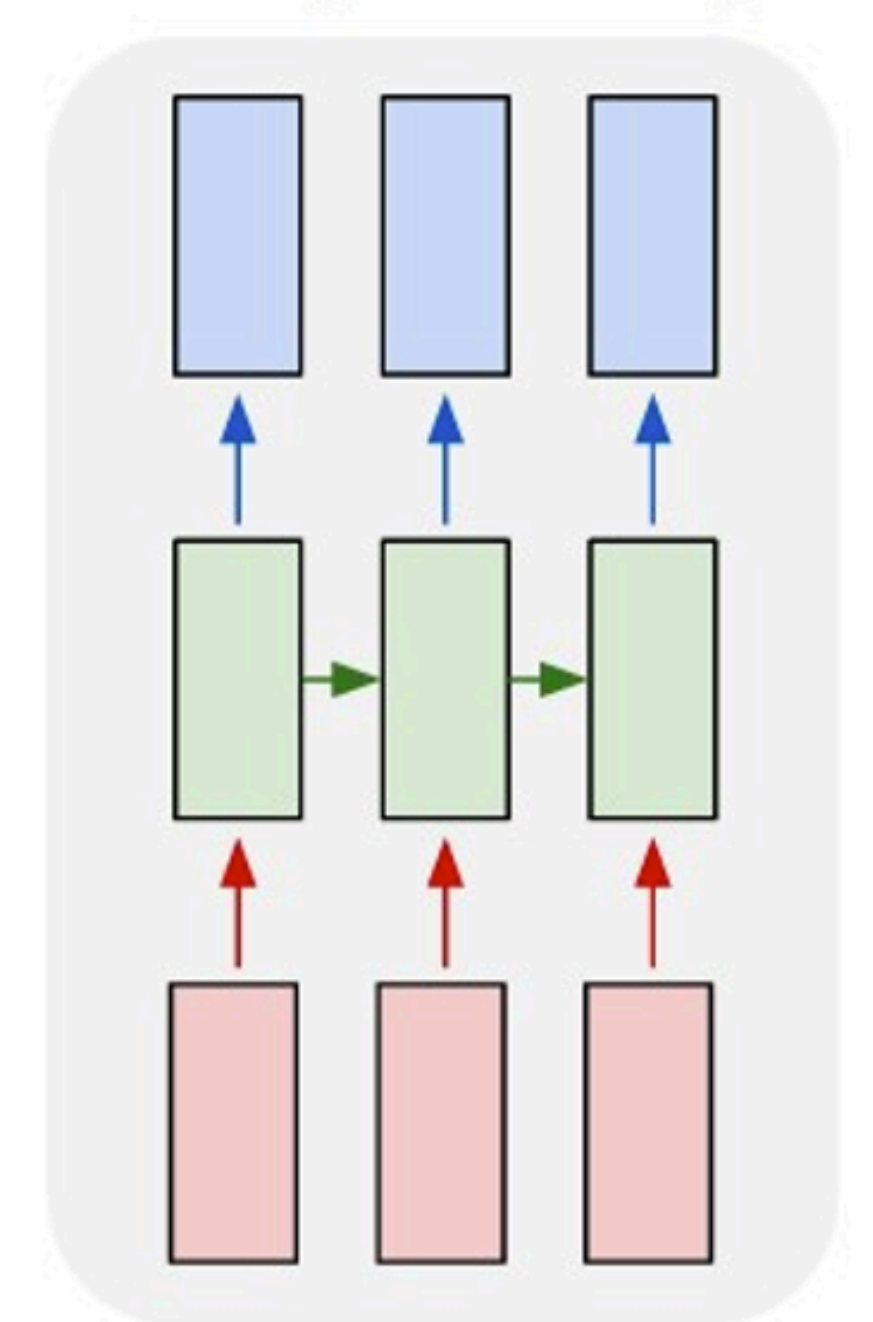
Text
Classification

many to many



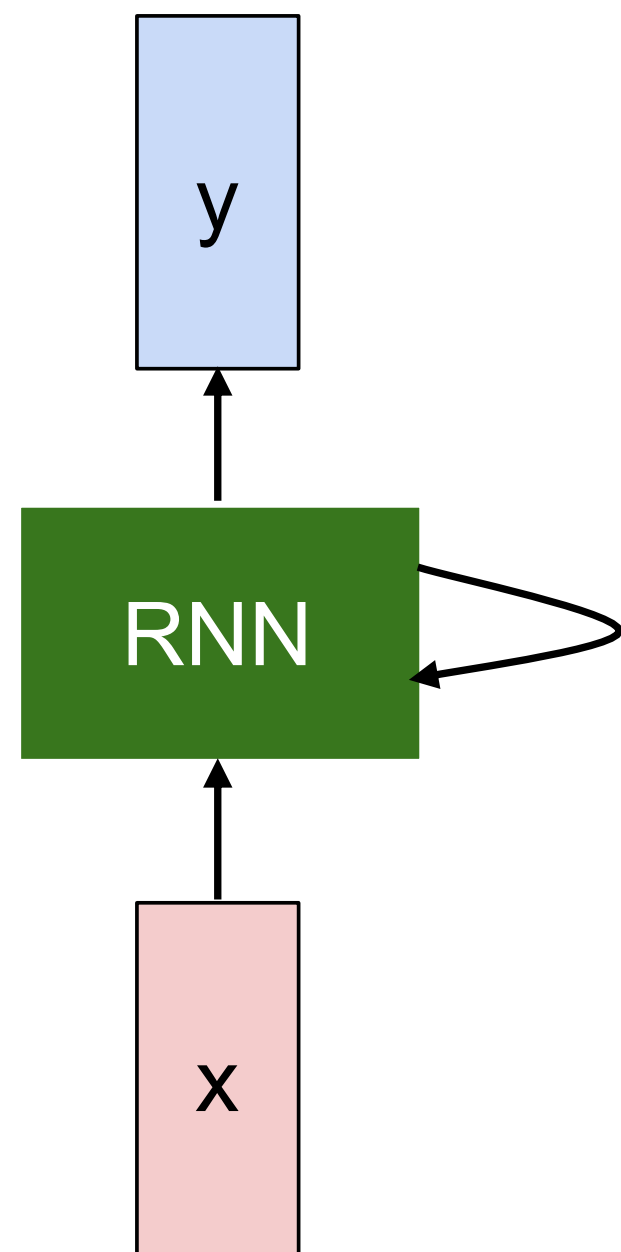
Neural Machine
Translation

many to many



Sequence
Tagging

Simple (vanilla) RNNs



$\mathbf{h}_0 \in \mathbb{R}^d$ is an initial state

function with weights W

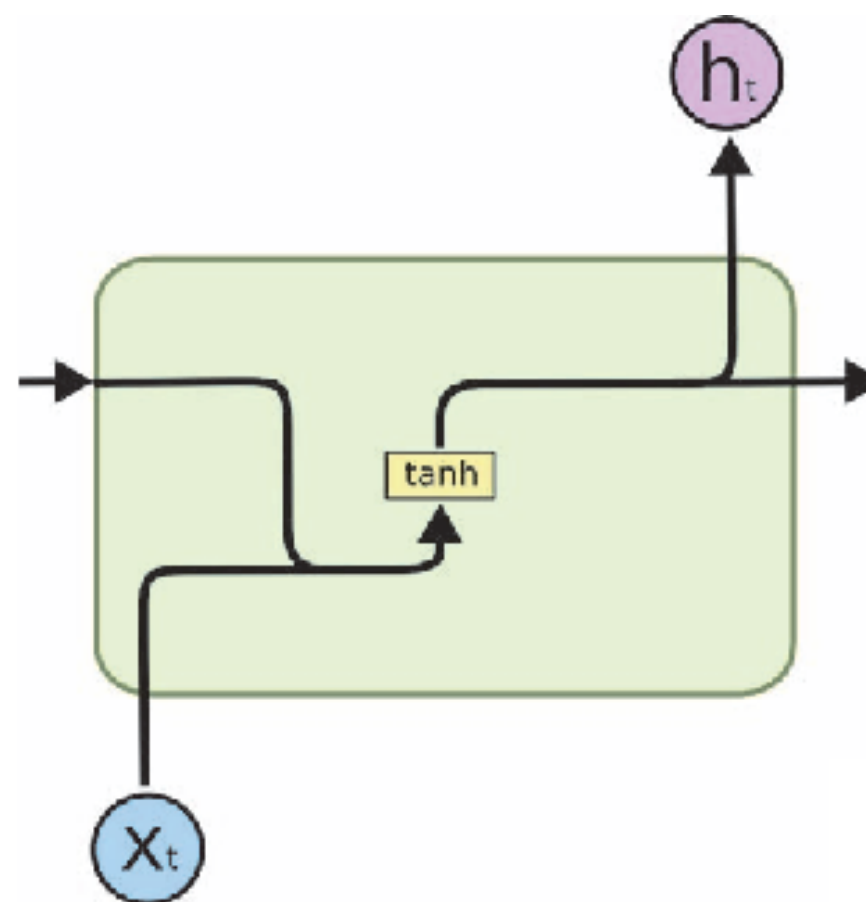
$$\mathbf{h}_t = f_W(\mathbf{h}_{t-1}, \mathbf{x}_t) \in \mathbb{R}^d$$

new state old state input at time t

$\mathbf{h}_t$: hidden states which store information from $\mathbf{x}_1$ to $\mathbf{x}_t$

Output label for each time step: Denote $\hat{\mathbf{y}}_t = \text{softmax}(\mathbf{W}_o \mathbf{h}_t)$, $\mathbf{W}_o \in \mathbb{R}^{|L| \times d}$

Simple (vanilla) RNNs:



$$\mathbf{h}_t = g(\mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t + \mathbf{b}) \in \mathbb{R}^d$$

g : nonlinearity (e.g. tanh),

$$\mathbf{W}_h \in \mathbb{R}^{d \times d}, \mathbf{W}_x \in \mathbb{R}^{d \times d_{in}}, \mathbf{b} \in \mathbb{R}^d$$

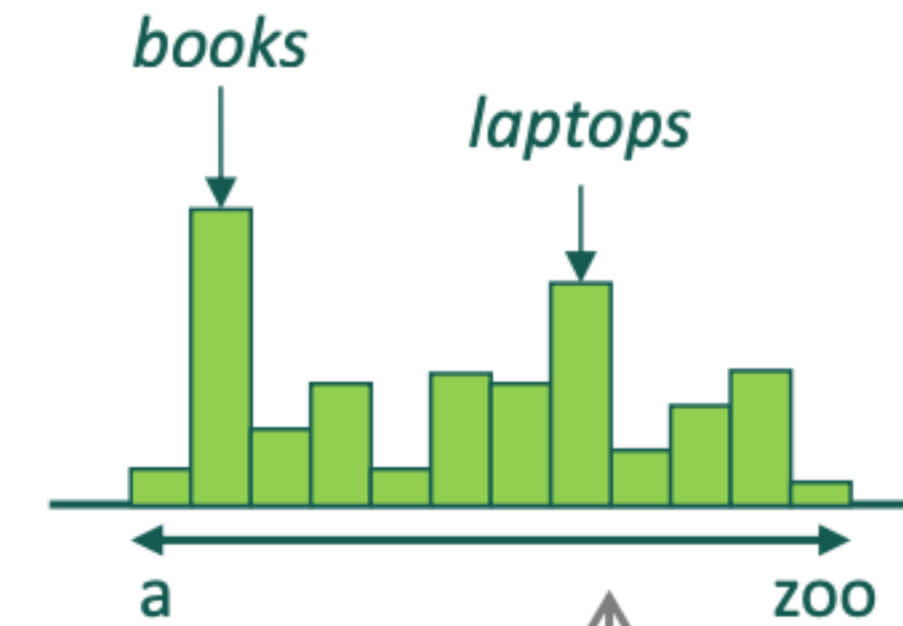
RNN Language Model

output distribution

$$\hat{y}^{(t)} = \text{softmax}(U h^{(t)} + b_2) \in \mathbb{R}^{|V|}$$

Output label size: $|V|$

$$\hat{y}^{(4)} = P(x^{(5)} | \text{the students opened their})$$



hidden states

$$h^{(t)} = \sigma(W_h h^{(t-1)} + W_e e^{(t)} + b_1)$$

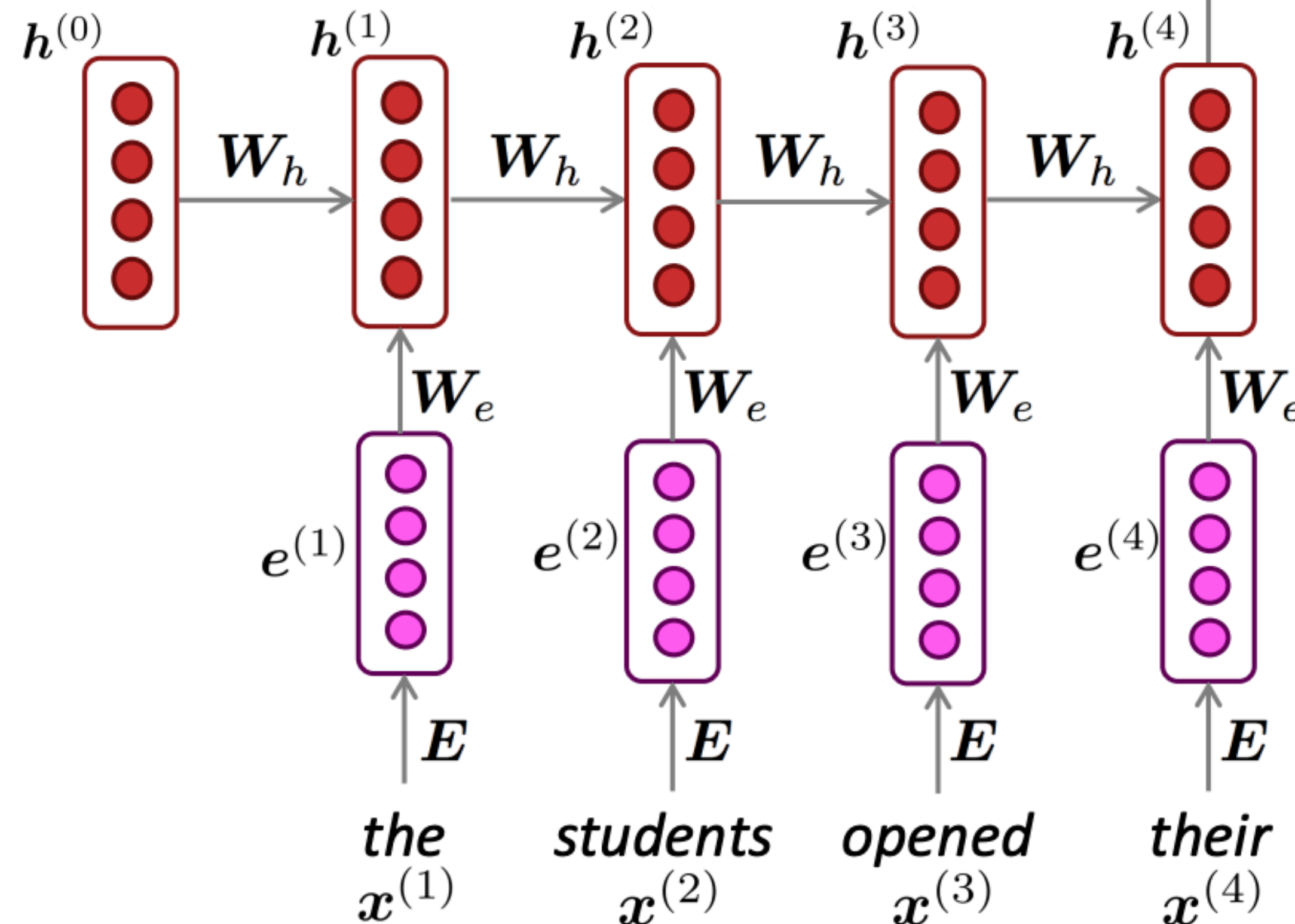
$h^{(0)}$ is the initial hidden state

word embeddings

$$e^{(t)} = E x^{(t)}$$

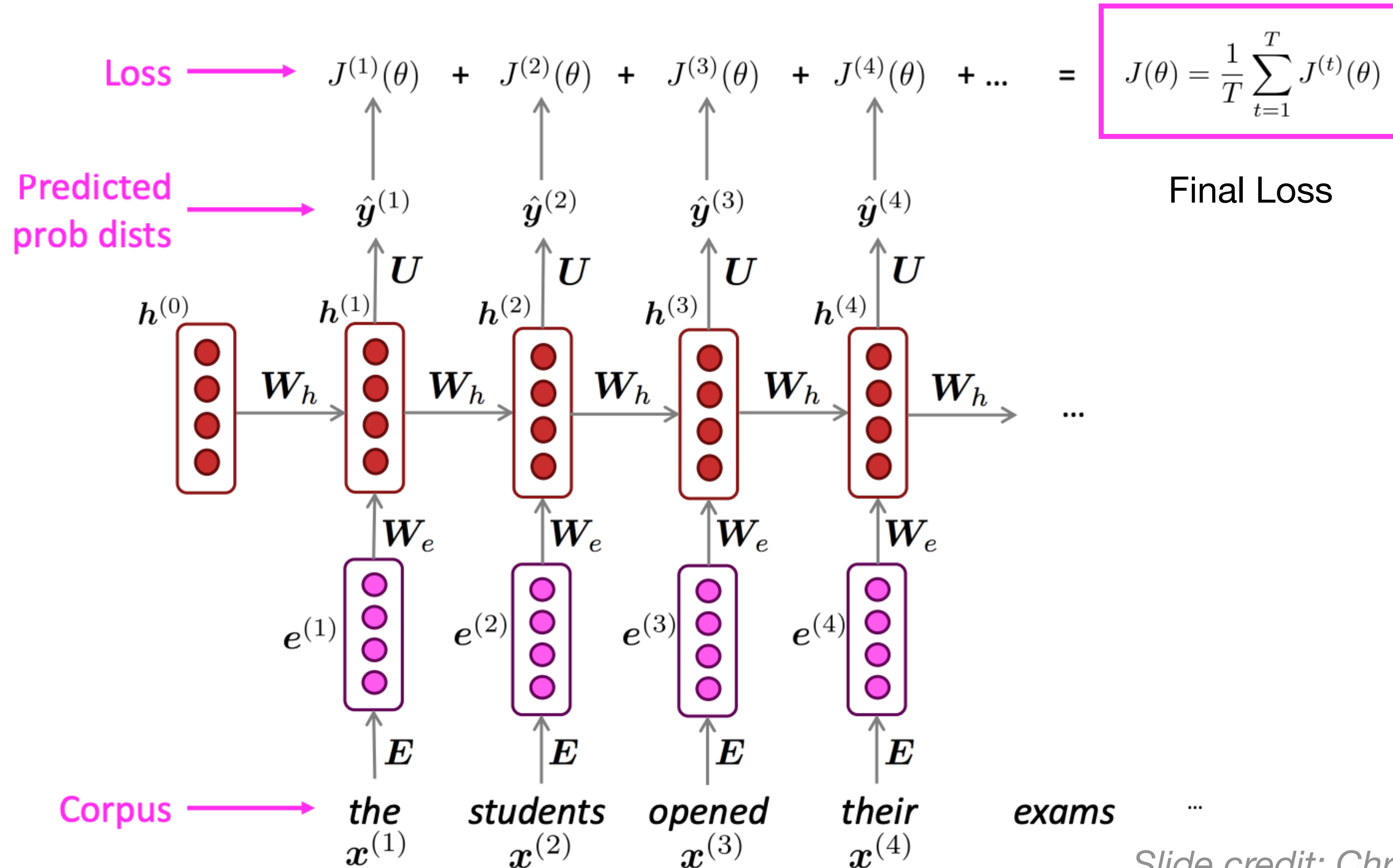
words / one-hot vectors

$$x^{(t)} \in \mathbb{R}^{|V|}$$



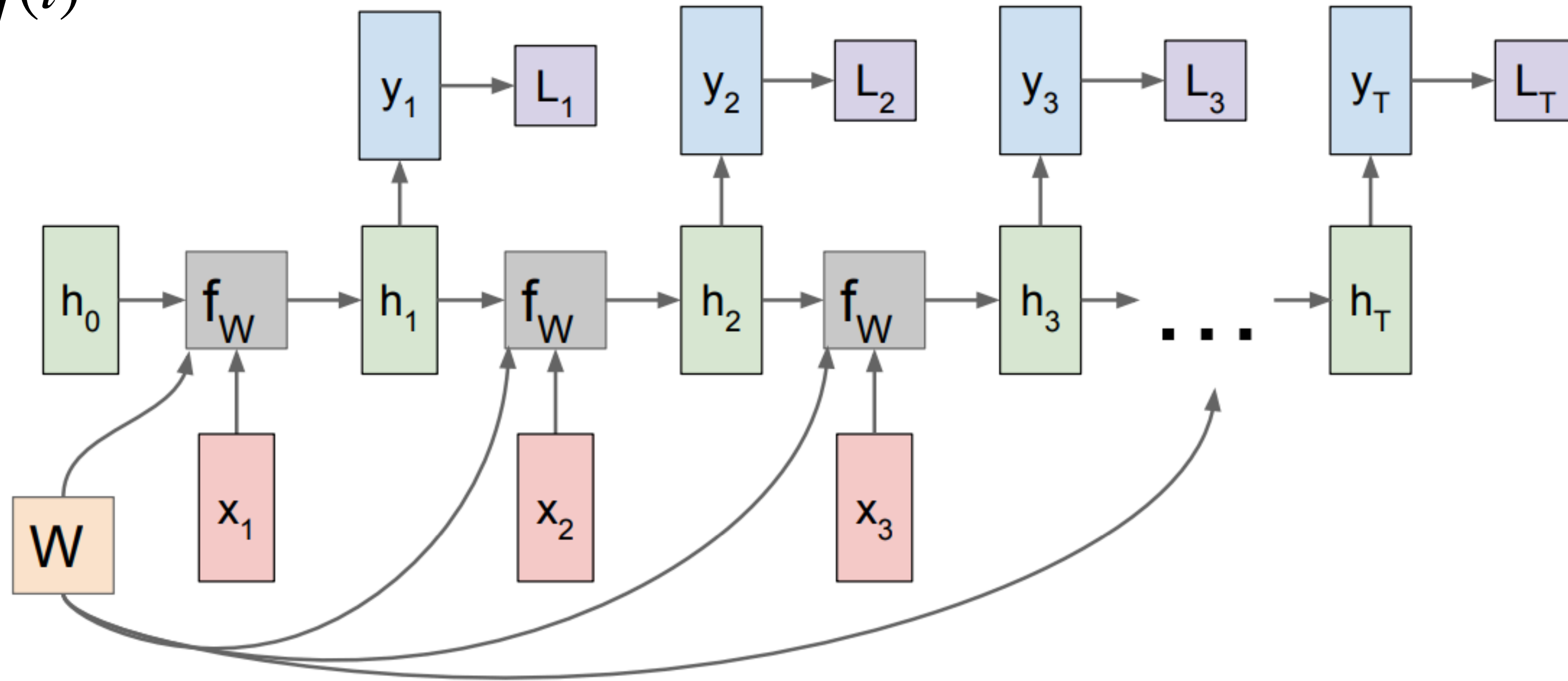
Use word embeddings

Training an RNN Language Model



Unrolled Computational Graph

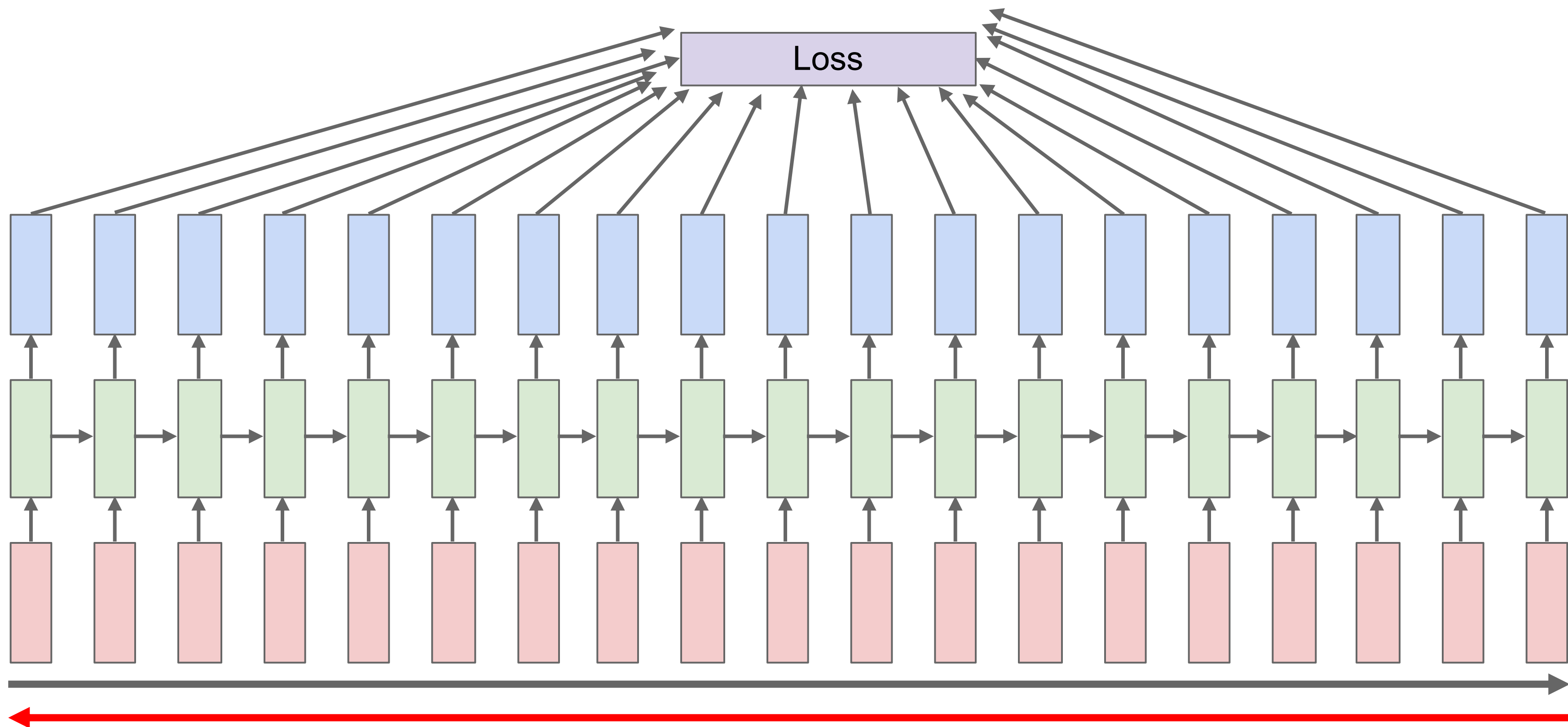
$$L_t = J^{(t)}$$



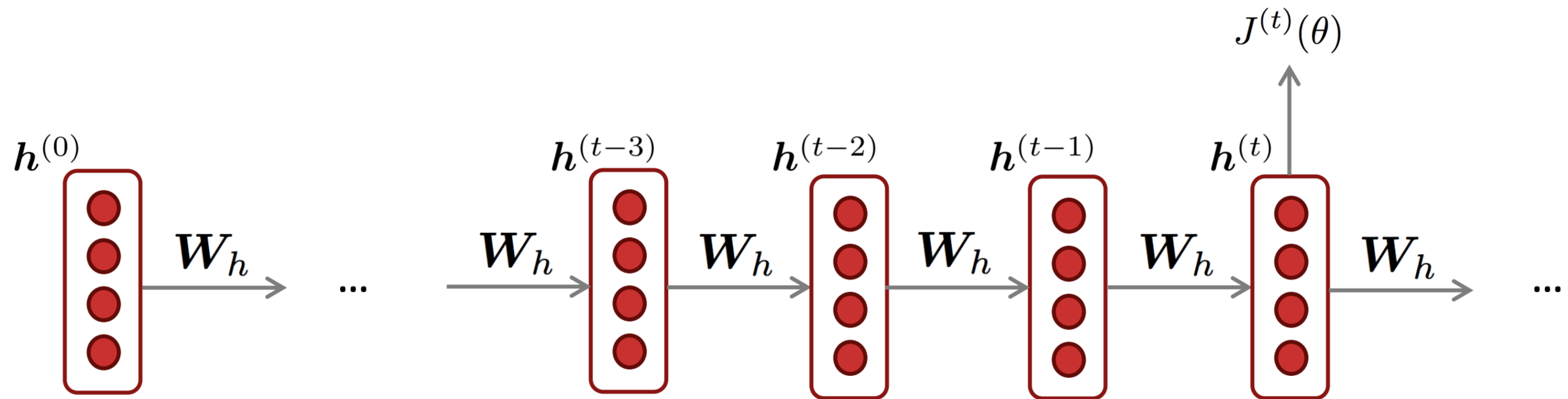
- Forward Pass: Compute losses for each time step and sum it together
- Backward Pass: Backpropagation through time (BPTT)

Backpropagation through time

Forward through entire sequence to compute loss, then
backward through entire sequence to compute gradient



Let's consider the gradient wrt the weight matrix



$$\frac{\partial J}{\partial \mathbf{W}_h} = -\frac{1}{n} \sum_{t=1}^n \frac{\partial J^{(t)}}{\partial \mathbf{W}_h}$$

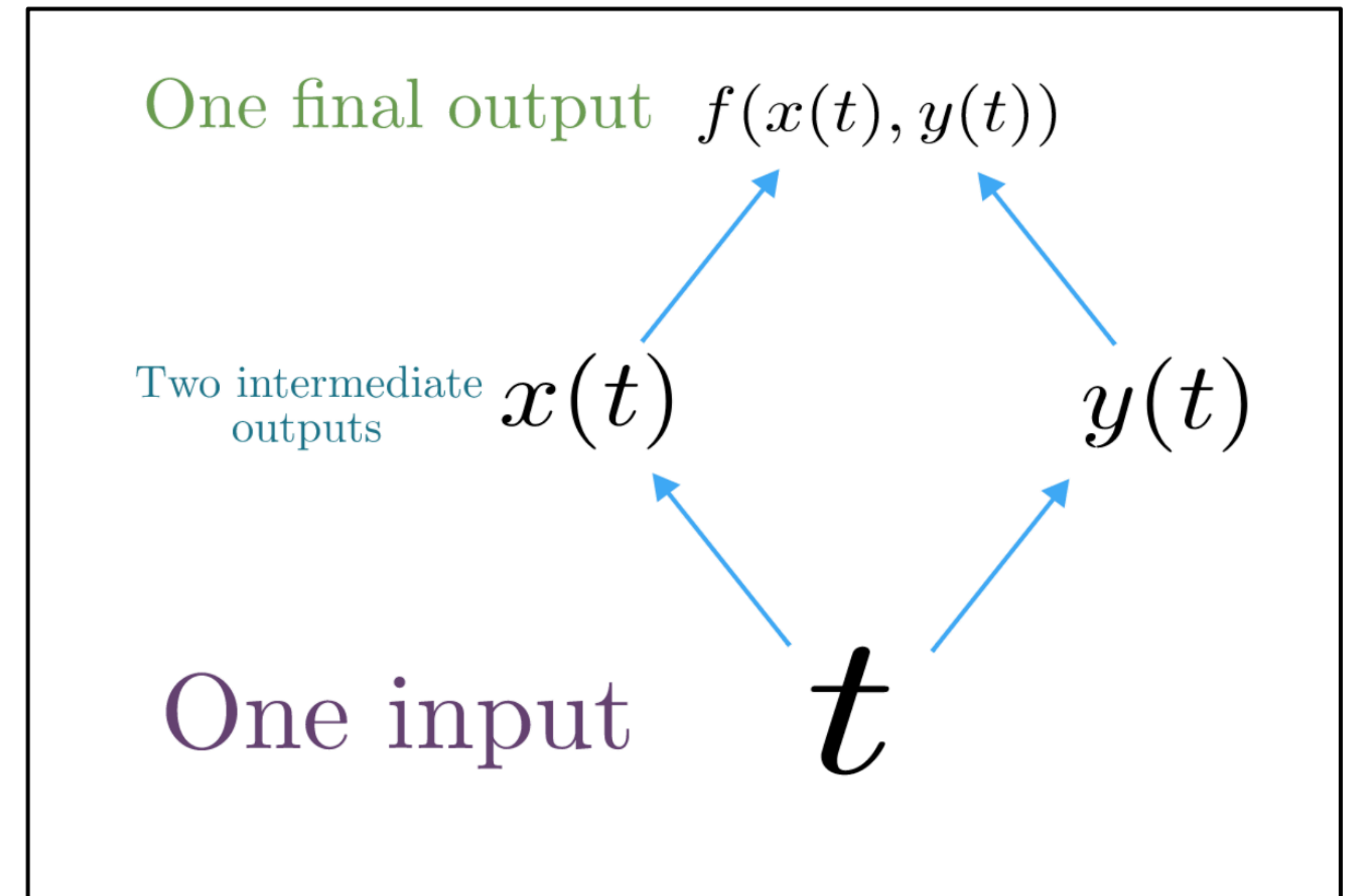
$$\frac{\partial J^{(t)}}{\partial \mathbf{W}_h} = \sum_{i=1}^t \frac{\partial J^{(i)}}{\partial \mathbf{W}_h} \Big|_{(i)}$$

Gradient wrt a repeated weight is the sum of the gradient wrt each time it appears

Recall: Gradient sum at branches

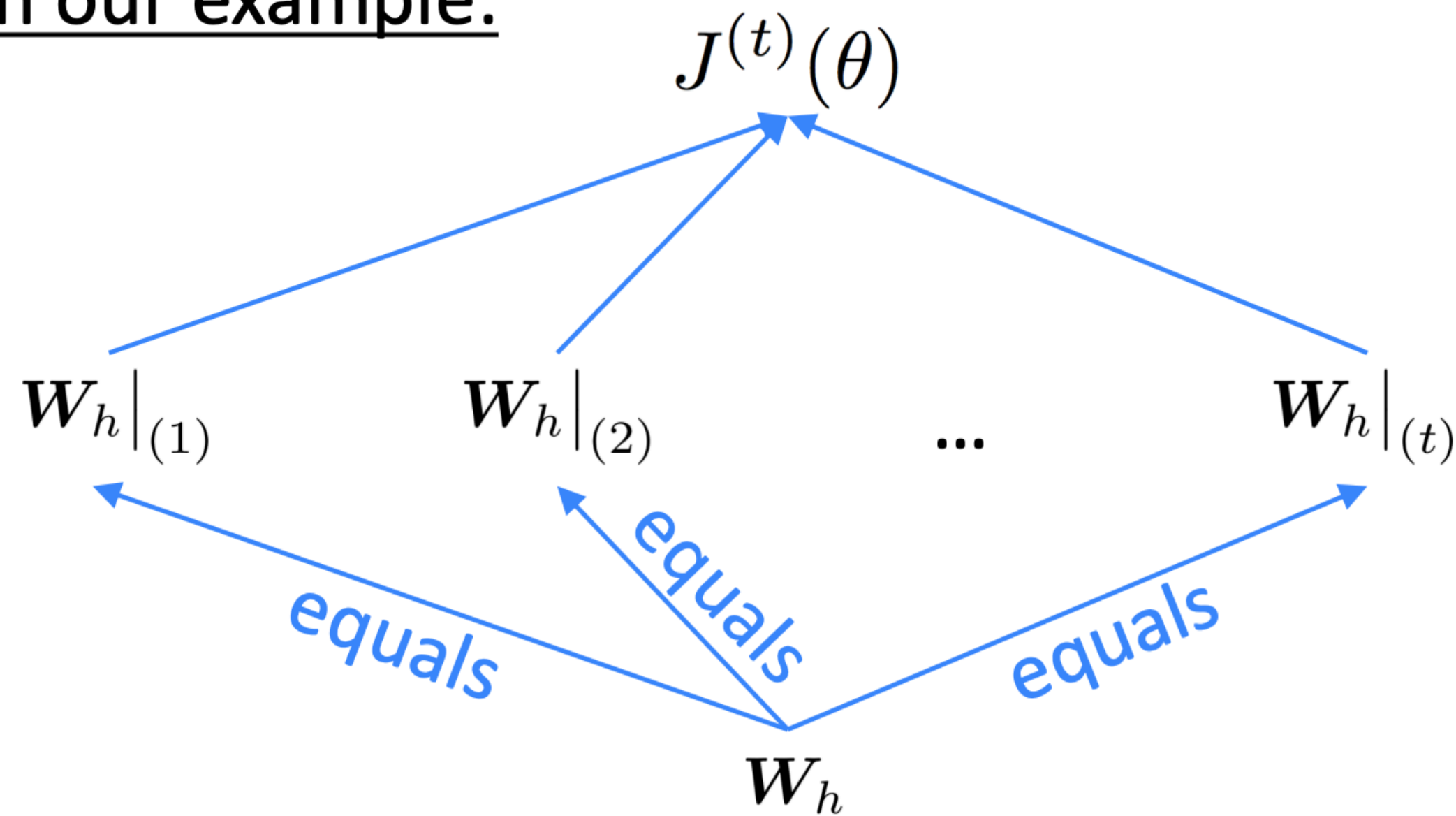
Multivariate Chain Rule

$$\underbrace{\frac{d}{dt} f(\mathbf{x}(t), \mathbf{y}(t))}_{\text{Derivative of composition function}} = \frac{\partial f}{\partial \mathbf{x}} \frac{d\mathbf{x}}{dt} + \frac{\partial f}{\partial \mathbf{y}} \frac{d\mathbf{y}}{dt}$$



Recall: Gradient sum at branches

In our example:



Apply the multivariable chain rule:

$$\begin{aligned} \frac{\partial J^{(t)}}{\partial \mathbf{w}_h} &= \sum_{i=1}^t \frac{\partial J^{(t)}}{\partial \mathbf{w}_h} \Big|_{(i)} \overset{=1}{\boxed{\frac{\partial \mathbf{w}_h|_{(i)}}{\partial \mathbf{w}_h}}} \\ &= \sum_{i=1}^t \frac{\partial J^{(t)}}{\partial \mathbf{w}_h} \Big|_{(i)} \end{aligned}$$

$$\frac{\partial J^{(t)}}{\partial \mathbf{W}_h} = \sum_{i=1}^t \frac{\partial J^{(t)}}{\partial \mathbf{W}_h} \Big|_{(i)}$$

BPTT: Example for $t=3$

$$L_t = J^{(t)}$$

$$\mathbf{h}_1 = g(\mathbf{W}_h \mathbf{h}_0 + \mathbf{W}_x \mathbf{x}_1 + \mathbf{b})$$

$$\mathbf{h}_2 = g(\mathbf{W}_h \mathbf{h}_1 + \mathbf{W}_x \mathbf{x}_2 + \mathbf{b})$$

$$\mathbf{h}_3 = g(\mathbf{W}_h \mathbf{h}_2 + \mathbf{W}_x \mathbf{x}_3 + \mathbf{b})$$

$$L_3 = -\log \hat{y}_3(w_4)$$

If i and t are far away, the gradients are likely to grow/shrink exponentially (called the **exploding or vanishing gradient** problem)

You should know how to compute: $\frac{\partial L_3}{\partial \mathbf{h}_3}$

$$\frac{\partial L_3}{\partial \mathbf{W}_h} = \frac{\partial L_3}{\partial \mathbf{h}_3} \frac{\partial \mathbf{h}_3}{\partial \mathbf{W}_h} + \frac{\partial L_3}{\partial \mathbf{h}_3} \frac{\partial \mathbf{h}_3}{\partial \mathbf{h}_2} \frac{\partial \mathbf{h}_2}{\partial \mathbf{W}_h} + \frac{\partial L_3}{\partial \mathbf{h}_3} \frac{\partial \mathbf{h}_3}{\partial \mathbf{h}_2} \frac{\partial \mathbf{h}_2}{\partial \mathbf{h}_1} \frac{\partial \mathbf{h}_1}{\partial \mathbf{W}_h}$$

$$\frac{\partial L}{\partial \mathbf{W}_h} = -\frac{1}{n} \sum_{t=1}^n \sum_{i=1}^t \frac{\partial L_t}{\partial \mathbf{h}_t} \left(\prod_{j=i+1}^t \frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}} \right) \frac{\partial \mathbf{h}_i}{\partial \mathbf{W}_h}$$

Exploding and vanishing gradients

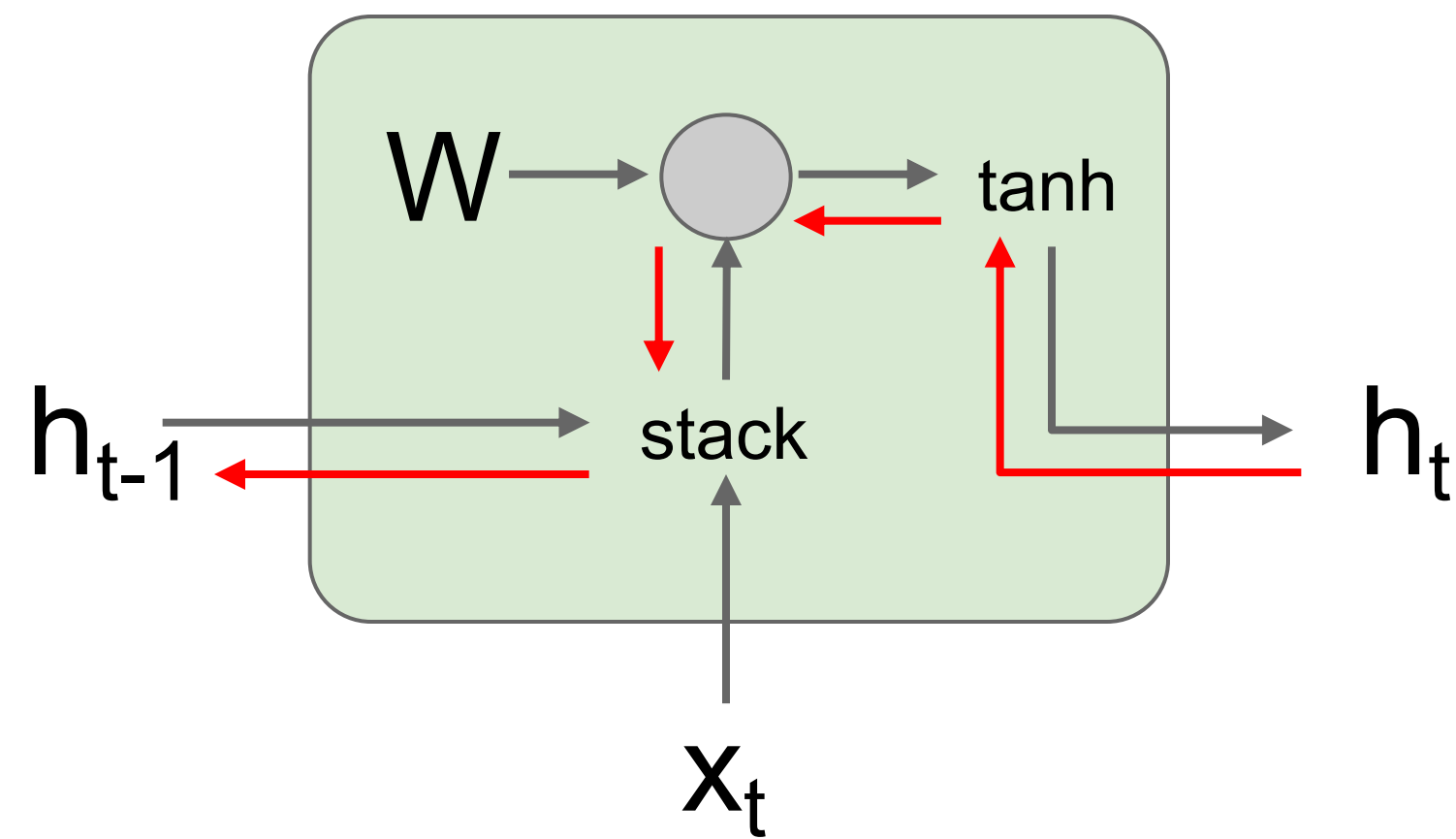
Vanishing/exploding gradients ^(advanced)

- Consider the gradient of L_t at step t , with respect to the hidden state $\mathbf{h}_k$ at some previous step k ($k < t$):

$$\begin{aligned}\frac{\partial L_t}{\partial \mathbf{h}_k} &= \frac{\partial L_t}{\partial \mathbf{h}_t} \left(\prod_{t \geq j > k} \frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}} \right) \\ &= \frac{\partial L_t}{\partial \mathbf{h}_t} \times \prod_{t \geq j > k} \left(\text{diag} \left(g'(\mathbf{W}\mathbf{h}_{j-1} + \mathbf{U}\mathbf{x}_j + \mathbf{b}) \right) \mathbf{W} \right)\end{aligned}$$

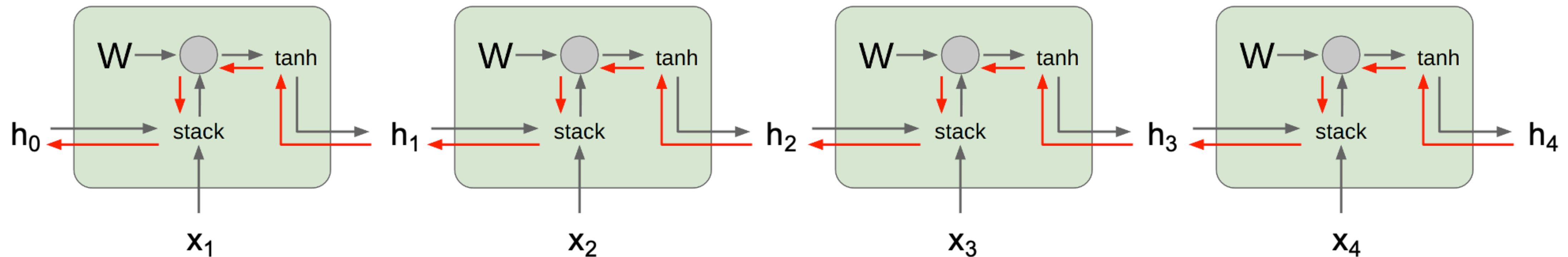
- (Pascanu et al, 2013) showed that if the largest eigenvalue of $\mathbf{W}$ is less than 1 for $g = \tanh$, then the gradient will shrink exponentially. This problem is called **vanishing gradients**.
- In contrast, if the gradients are getting too large, it is called **exploding gradients**.

Gradient flow through Vanilla RNN cell



$$\begin{aligned} h_t &= \tanh(W_{hh}h_{t-1} + W_{hx}x_t) \\ &= \tanh \left((W_{hh} \quad W_{hx}) \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} \right) \\ &= \tanh \left(W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} \right) \end{aligned}$$

Exploding and Vanishing Gradients



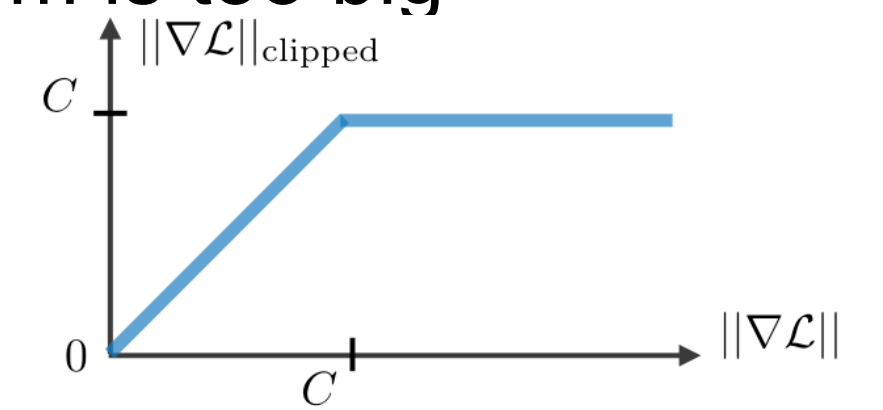
Computing gradient of h_0
involves many factors of W
(and repeated tanh)

Largest singular value > 1 :
Exploding gradients



Gradient clipping:
Scale gradient if its norm is too big

```
grad_norm = np.sum(grad * grad)
if grad_norm > threshold:
    grad *= (threshold / grad_norm)
```



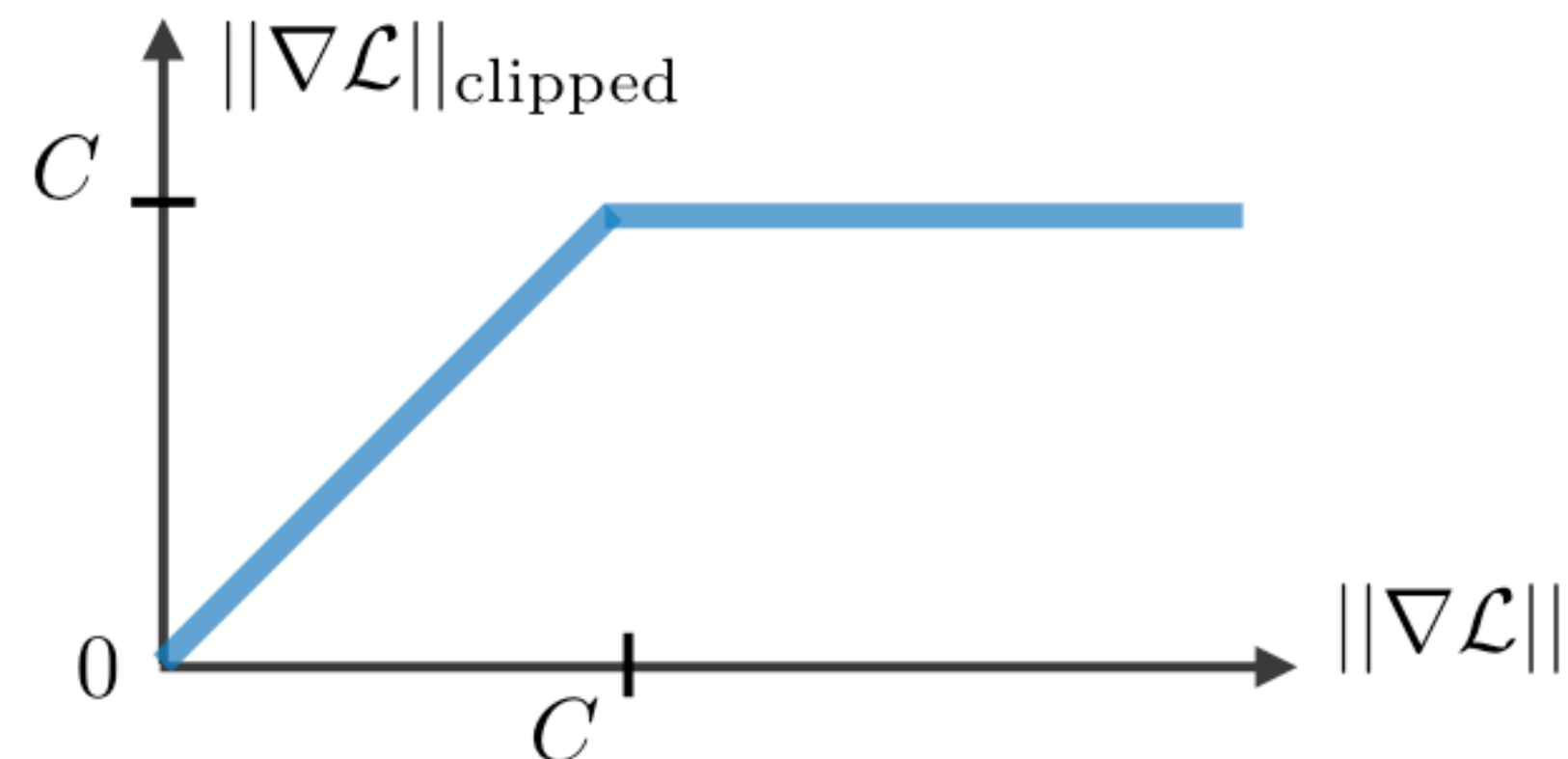
Largest singular value < 1 :
Vanishing gradients

Why is exploding gradient a problem?

- Gradients become too big and we take a very large step in SGD.
- **Solution:** Gradient clipping — if the norm of the gradient is greater than some threshold, scale it down before applying SGD update.

Algorithm 1 Pseudo-code for norm clipping

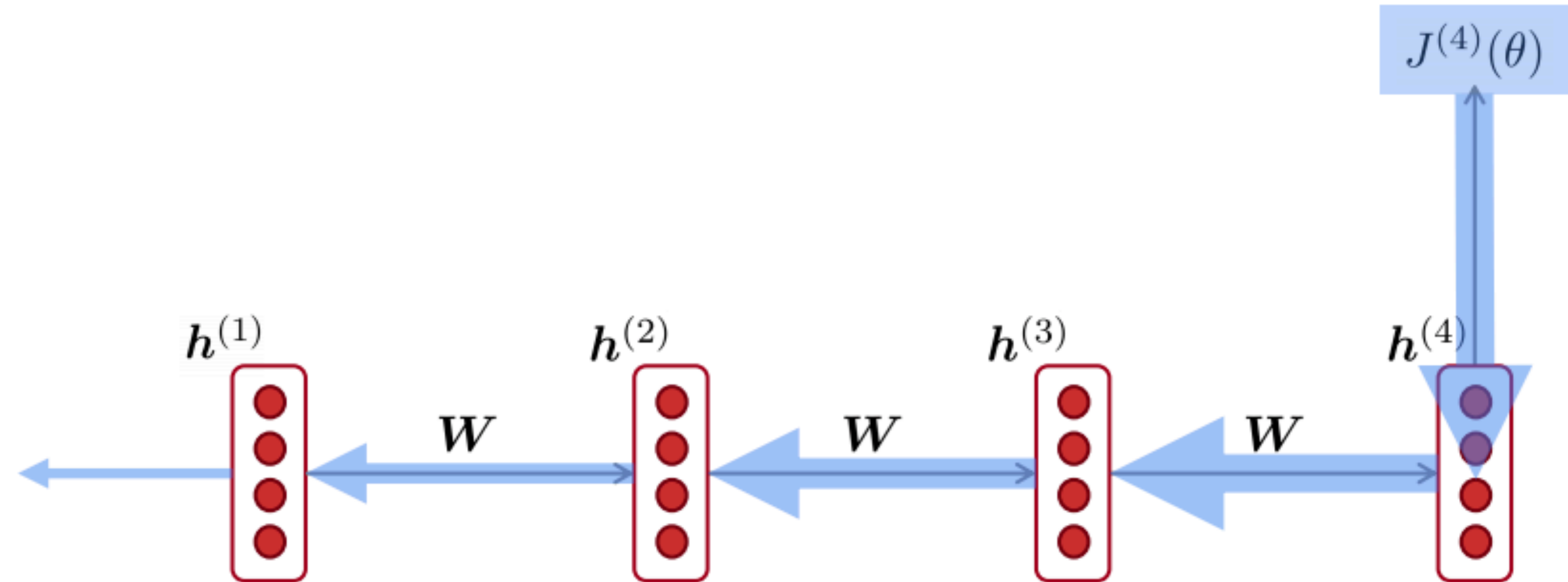
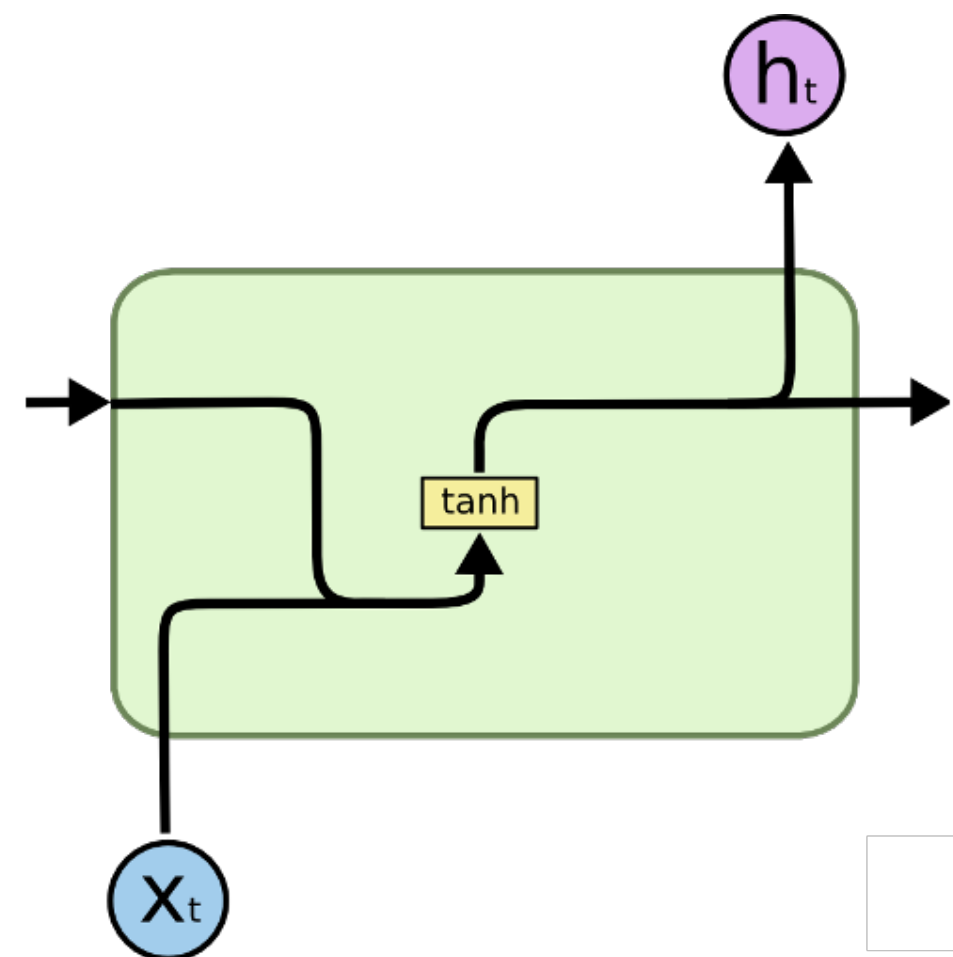
```
 $\hat{\mathbf{g}} \leftarrow \frac{\partial \mathcal{E}}{\partial \theta}$   
if  $\|\hat{\mathbf{g}}\| \geq \text{threshold}$  then  
     $\hat{\mathbf{g}} \leftarrow \frac{\text{threshold}}{\|\hat{\mathbf{g}}\|} \hat{\mathbf{g}}$   
end if
```



Vanishing gradients

Simple RNN

$$h_t = \tanh \left(W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} \right)$$

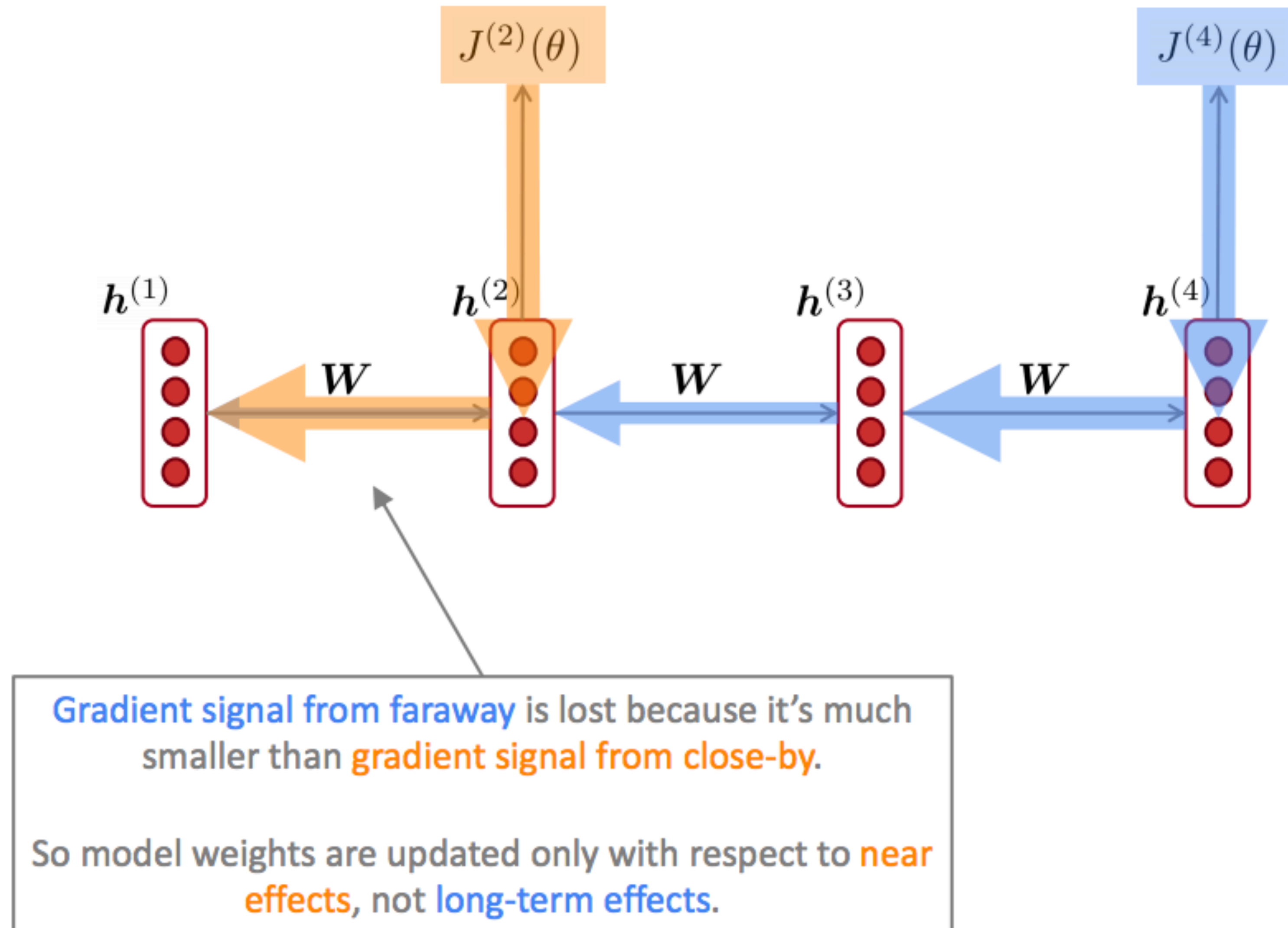


$$\frac{\partial J^{(4)}}{\partial h^{(1)}} = \frac{\partial h^{(2)}}{\partial h^{(1)}} \times \frac{\partial h^{(3)}}{\partial h^{(2)}} \times \frac{\partial h^{(4)}}{\partial h^{(3)}} \times \frac{\partial J^{(4)}}{\partial h^{(4)}}$$

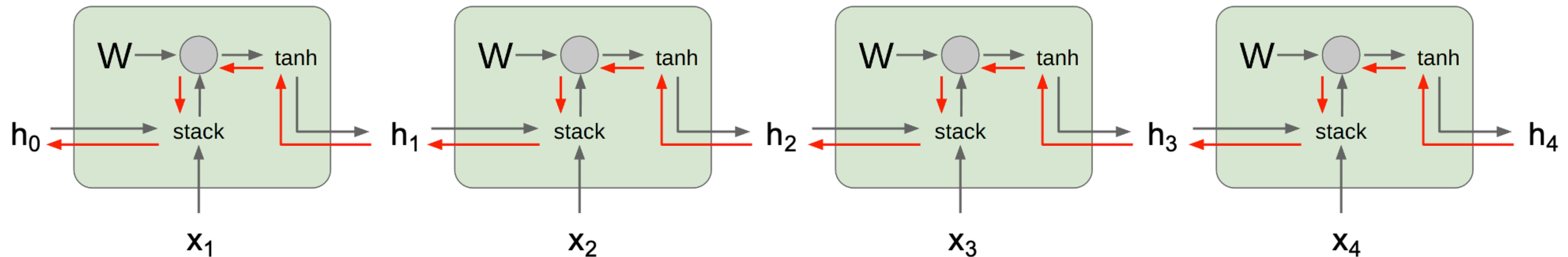
What happens if these are small?

Vanishing gradient problem:
When these are small, the gradient signal gets smaller and smaller as it backpropagates further

Vanishing Gradients



Exploding and Vanishing Gradients

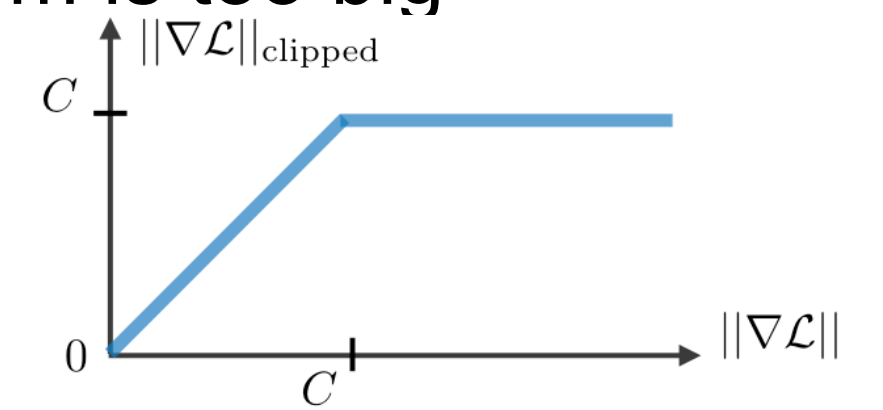


Computing gradient of h_0
involves many factors of W
(and repeated tanh)

Largest singular value > 1 :
Exploding gradients

Gradient clipping:
Scale gradient if its norm is too big

```
grad_norm = np.sum(grad * grad)
if grad_norm > threshold:
    grad *= (threshold / grad_norm)
```



Largest singular value < 1 :
Vanishing gradients

Change RNN architecture

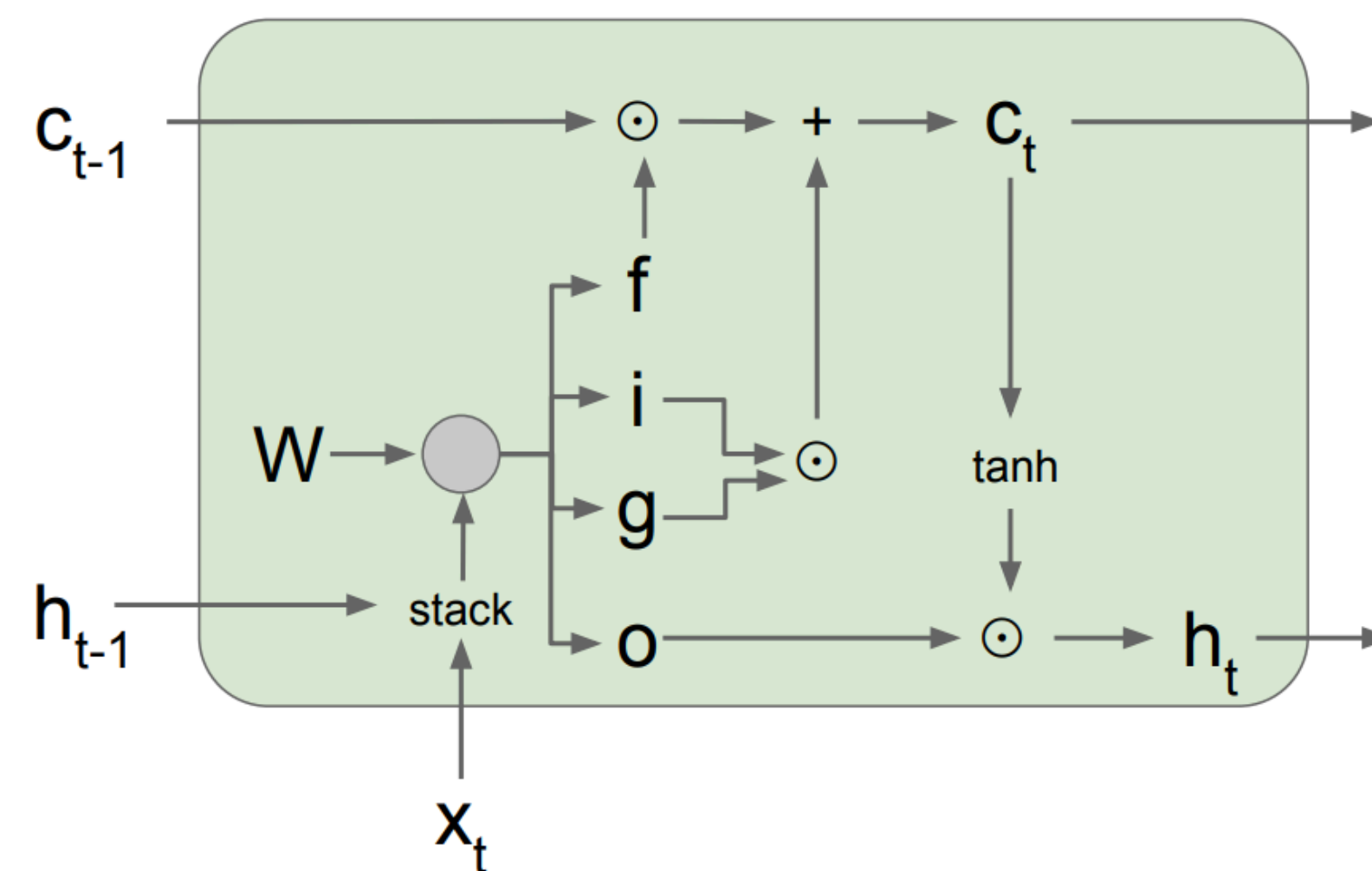
Different RNN cells

Long Short-term Memory (LSTM)

- A type of RNN proposed by Hochreiter and Schmidhuber in 1997 as a solution to the vanishing gradients problem
- Work extremely well in practice
- **Basic idea:** turning multiplication into addition
- Use “gates” to control how much information to add/erase

$$\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t) \in \mathbb{R}^d$$

- At each timestep, there is a hidden state $\mathbf{h}_t \in \mathbb{R}^d$ and also a cell state $\mathbf{c}_t \in \mathbb{R}^d$
 - $\mathbf{c}_t$ stores **long-term information**
 - We write/erase $\mathbf{c}_t$ after each step
 - We read $\mathbf{h}_t$ from $\mathbf{c}_t$



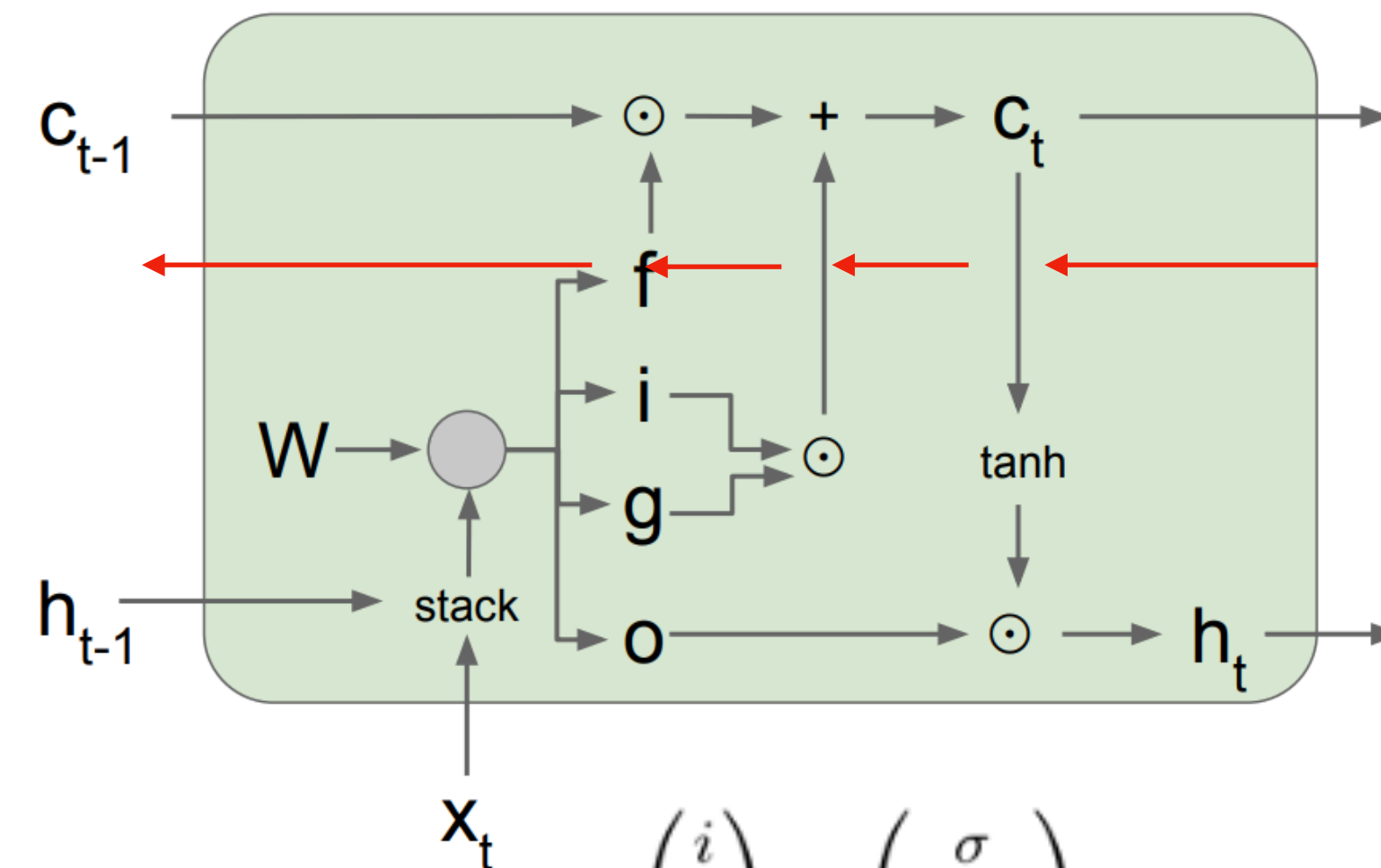
Long Short-term Memory (LSTM)

There are 4 gates:

- Input gate (how much to write):
 $\mathbf{i}_t = \sigma(\mathbf{W}^{(i)}\mathbf{h}_{t-1} + \mathbf{U}^{(i)}\mathbf{x}_t + \mathbf{b}^{(i)}) \in \mathbb{R}^d$
- Forget gate (how much to erase):
 $\mathbf{f}_t = \sigma(\mathbf{W}^{(f)}\mathbf{h}_{t-1} + \mathbf{U}^{(f)}\mathbf{x}_t + \mathbf{b}^{(f)}) \in \mathbb{R}^d$
- Output gate (how much to reveal):
 $\mathbf{o}_t = \sigma(\mathbf{W}^{(o)}\mathbf{h}_{t-1} + \mathbf{U}^{(o)}\mathbf{x}_t + \mathbf{b}^{(o)}) \in \mathbb{R}^d$
- New memory cell (what to write):
 $\mathbf{g}_t = \tanh(\mathbf{W}^{(c)}\mathbf{h}_{t-1} + \mathbf{U}^{(c)}\mathbf{x}_t + \mathbf{b}^{(c)}) \in \mathbb{R}^d$
- Final memory cell: $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t$
- Final hidden cell: $\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t)$

element-wise product

Backpropagation from c_t to c_{t-1}
 only element wise multiplication
 by f , no matrix multiply by W



$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

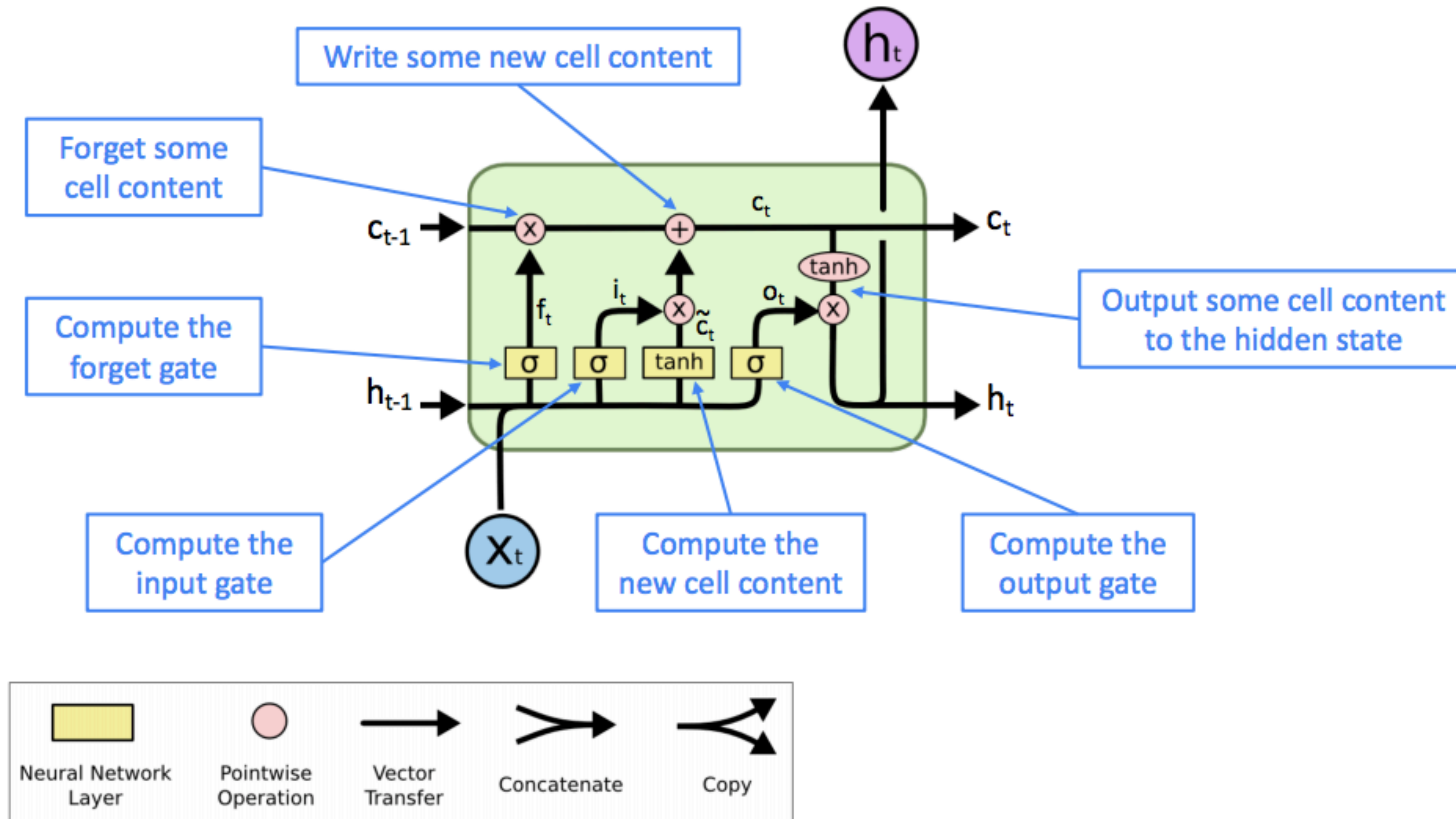
$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$

How many parameters in total? $4 \times (d^2 + dm + 1)$

LSTM cell intuitively

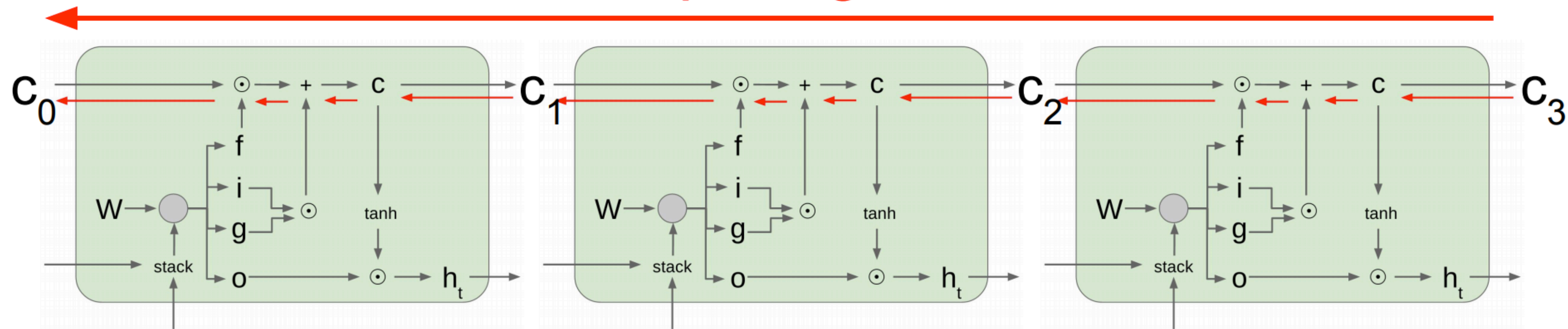
You can think of the LSTM equations visually like this:



<http://colah.github.io/posts/2015-08-Understanding-LSTMs/>

Long Short-term Memory (LSTM)

Uninterrupted gradient flow!



- LSTM doesn't guarantee that there is no vanishing/exploding gradient, but it does provide an easier way for the model to learn long-distance dependencies
- LSTMs were invented in 1997 but finally got working from 2013-2015.

Is the LSTM architecture optimal?

MUT1:

$$\begin{aligned} z &= \text{sigm}(W_{xz}x_t + b_z) \\ r &= \text{sigm}(W_{xr}x_t + W_{hr}h_t + b_r) \\ h_{t+1} &= \tanh(W_{hh}(r \odot h_t) + \tanh(x_t) + b_h) \odot z \\ &+ h_t \odot (1 - z) \end{aligned}$$

MUT2:

$$\begin{aligned} z &= \text{sigm}(W_{xz}x_t + W_{hz}h_t + b_z) \\ r &= \text{sigm}(x_t + W_{hr}h_t + b_r) \\ h_{t+1} &= \tanh(W_{hh}(r \odot h_t) + W_{xh}x_t + b_h) \odot z \\ &+ h_t \odot (1 - z) \end{aligned}$$

MUT3:

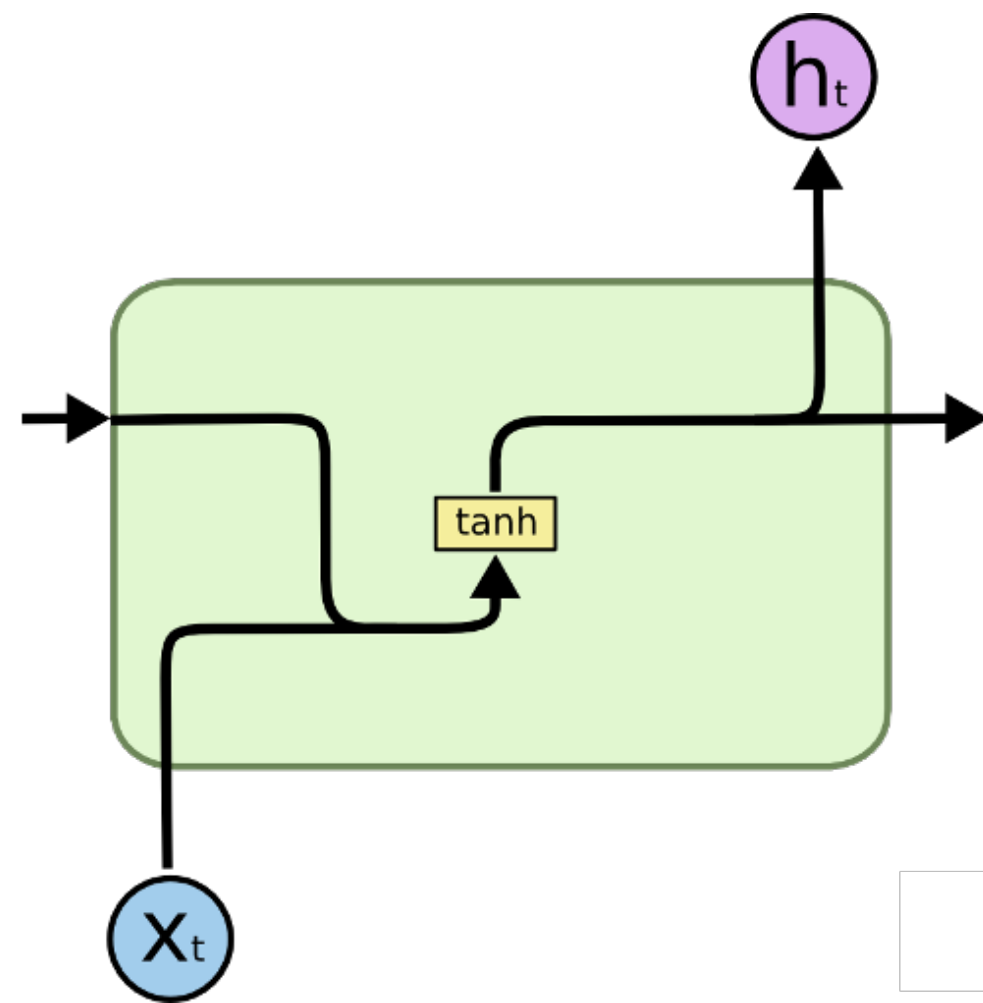
$$\begin{aligned} z &= \text{sigm}(W_{xz}x_t + W_{hz} \tanh(h_t) + b_z) \\ r &= \text{sigm}(W_{xr}x_t + W_{hr}h_t + b_r) \\ h_{t+1} &= \tanh(W_{hh}(r \odot h_t) + W_{xh}x_t + b_h) \odot z \\ &+ h_t \odot (1 - z) \end{aligned}$$

Arch.	Arith.	XML	PTB
Tanh	0.29493	0.32050	0.08782
LSTM	0.89228	0.42470	0.08912
LSTM-f	0.29292	0.23356	0.08808
LSTM-i	0.75109	0.41371	0.08662
LSTM-o	0.86747	0.42117	0.08933
LSTM-b	0.90163	0.44434	0.08952
GRU	0.89565	0.45963	0.09069
MUT1	0.92135	0.47483	0.08968
MUT2	0.89735	0.47324	0.09036
MUT3	0.90728	0.46478	0.09161

Arch.	5M-tst	10M-v	20M-v	20M-tst
Tanh	4.811	4.729	4.635	4.582 (97.7)
LSTM	4.699	4.511	4.437	4.399 (81.4)
LSTM-f	4.785	4.752	4.658	4.606 (100.8)
LSTM-i	4.755	4.558	4.480	4.444 (85.1)
LSTM-o	4.708	4.496	4.447	4.411 (82.3)
LSTM-b	4.698	4.437	4.423	4.380 (79.83)
GRU	4.684	4.554	4.559	4.519 (91.7)
MUT1	4.699	4.605	4.594	4.550 (94.6)
MUT2	4.707	4.539	4.538	4.503 (90.2)
MUT3	4.692	4.523	4.530	4.494 (89.47)

Simple RNN vs GRU vs LSTM

$$\mathbf{h}_t = g(\mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{U}\mathbf{x}_t + \mathbf{b})$$



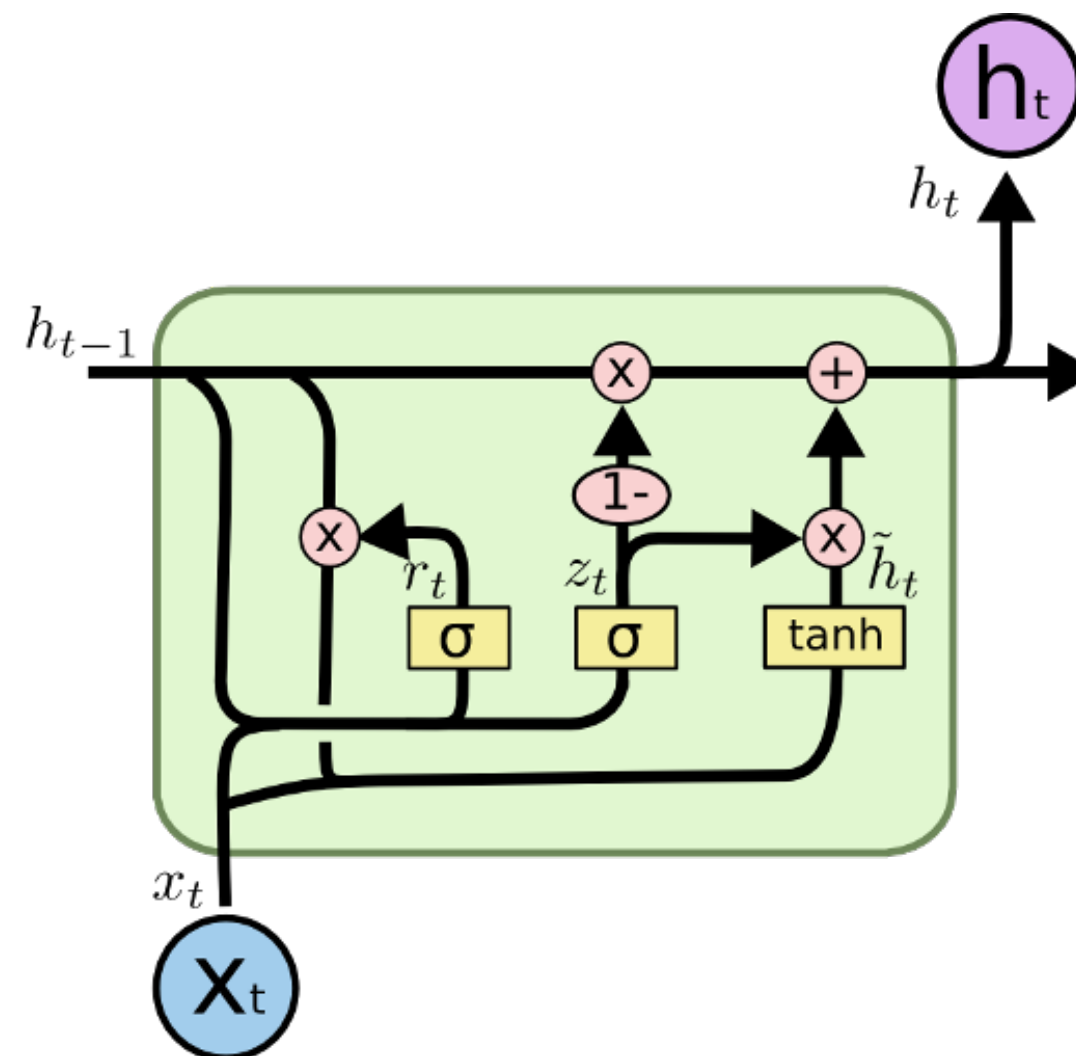
Simple RNN

$$\mathbf{r}_t = \sigma(\mathbf{W}^r\mathbf{h}_{t-1} + \mathbf{U}^r\mathbf{x}_t + \mathbf{b}^r)$$

$$\mathbf{z}_t = \sigma(\mathbf{W}^z\mathbf{h}_{t-1} + \mathbf{U}^z\mathbf{x}_t + \mathbf{b}^z)$$

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{U}\mathbf{x}_t + \mathbf{b})$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$$



GRU

$$\mathbf{i}_t = \sigma(\mathbf{W}^i\mathbf{h}_{t-1} + \mathbf{U}^i\mathbf{x}_t + \mathbf{b}^i)$$

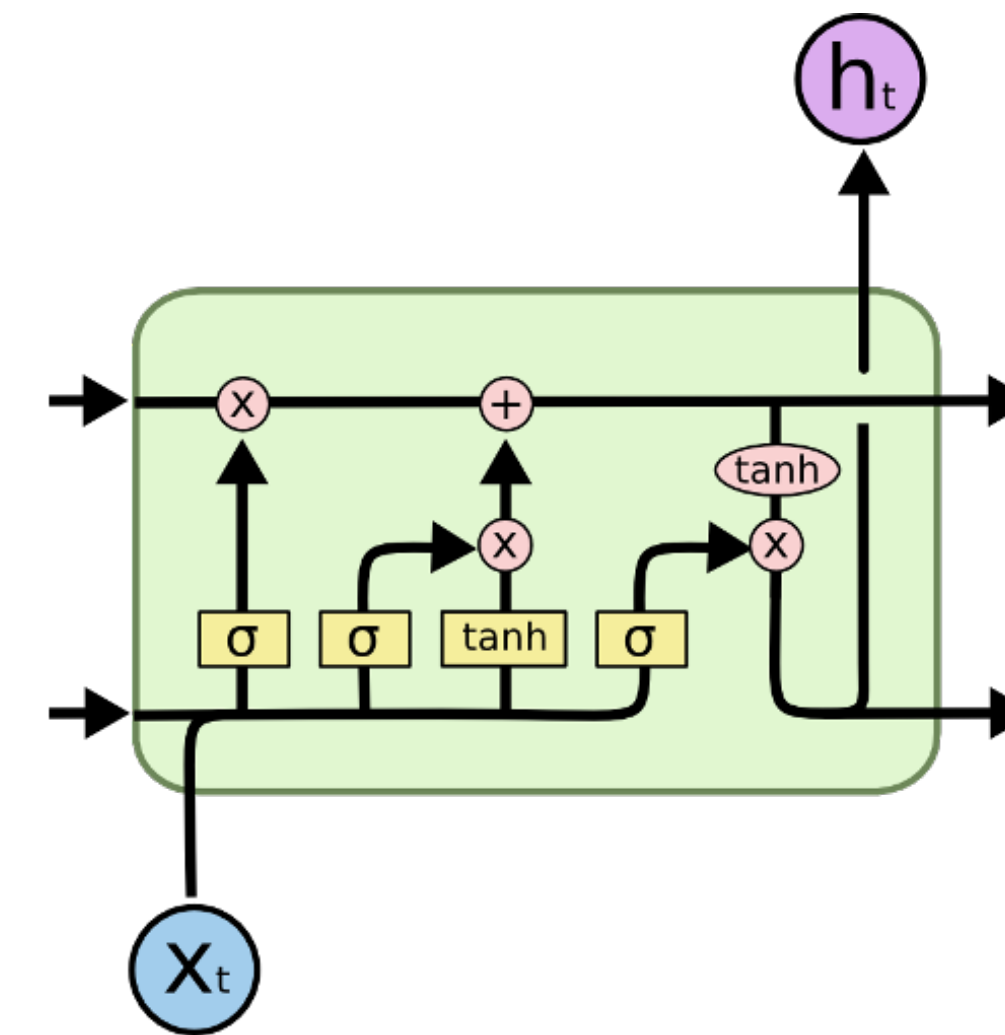
$$\mathbf{f}_t = \sigma(\mathbf{W}^f\mathbf{h}_{t-1} + \mathbf{U}^f\mathbf{x}_t + \mathbf{b}^f)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}^o\mathbf{h}_{t-1} + \mathbf{U}^o\mathbf{x}_t + \mathbf{b}^o)$$

$$\mathbf{g}_t = \tanh(\mathbf{W}^g\mathbf{h}_{t-1} + \mathbf{U}^g\mathbf{x}_t + \mathbf{b}^g)$$

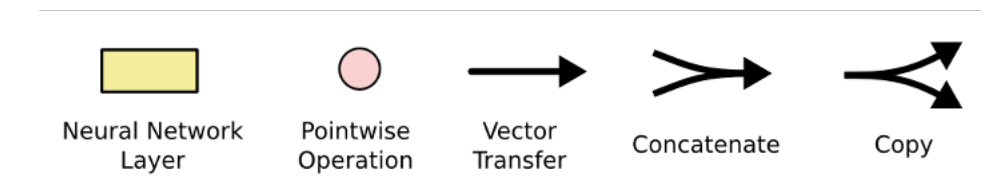
$$\mathbf{c}_t = \mathbf{c}_{t-1} \odot \mathbf{f}_t + \mathbf{g}_t \odot \mathbf{i}_t$$

$$\mathbf{h}_t = \tanh(\mathbf{c}_t) \odot \mathbf{o}_t$$



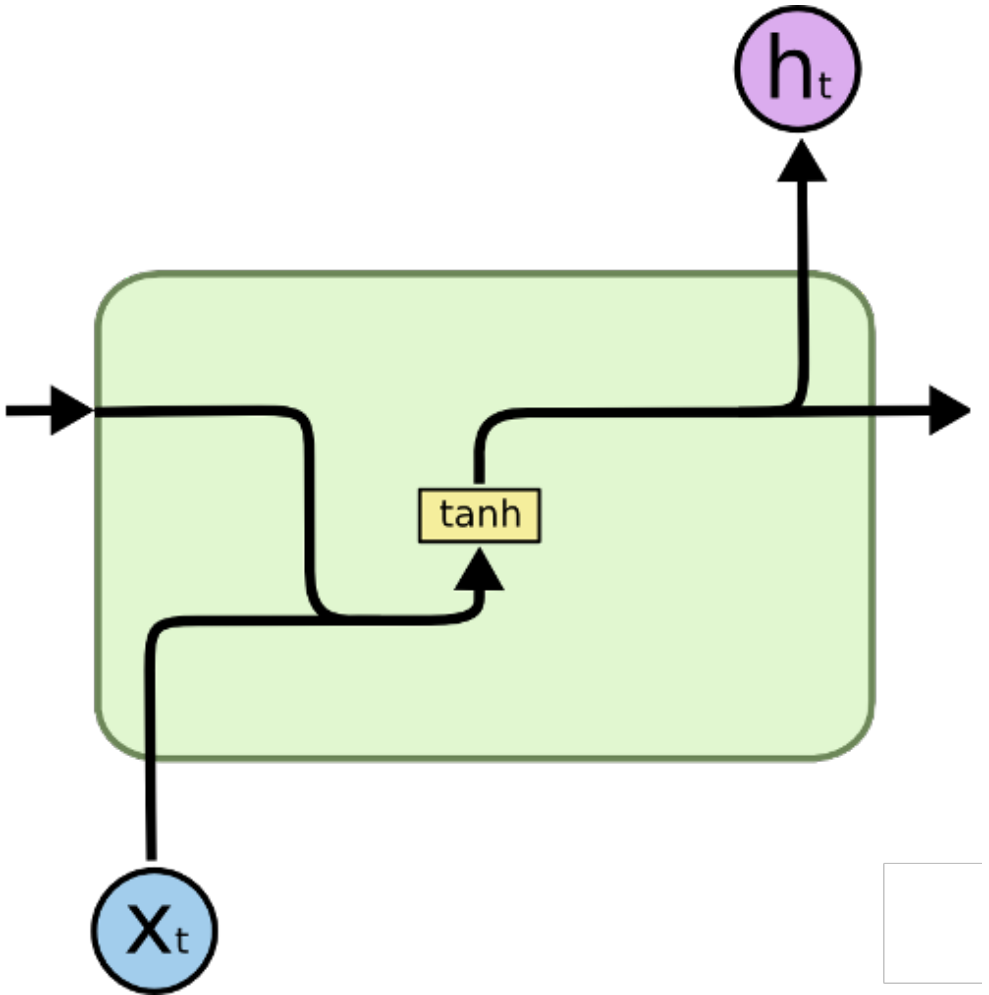
LSTM

<http://colah.github.io/posts/2015-08-Understanding-LSTMs/>



Simple RNN vs GRU vs LSTM

$$h_t = \tanh \left(W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} \right)$$

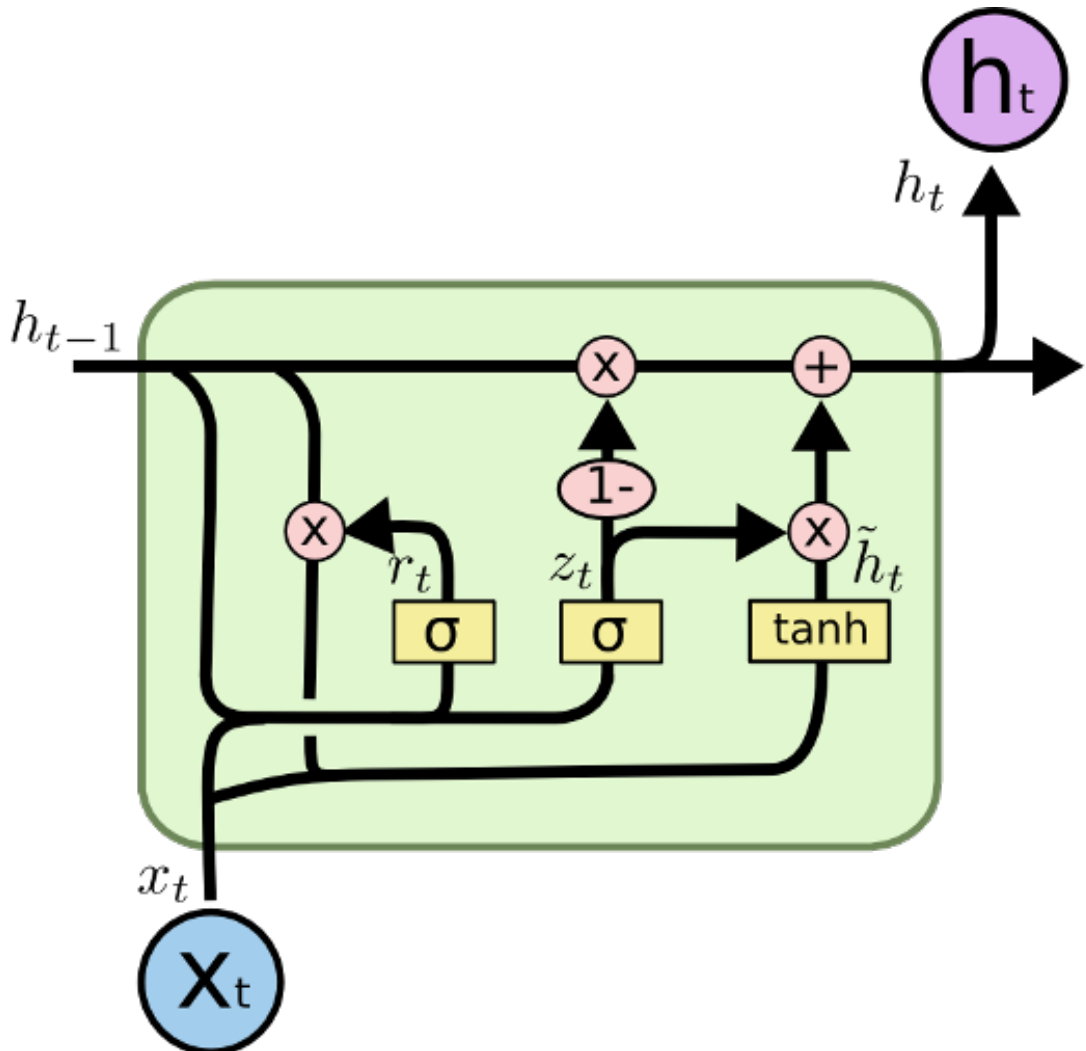


Simple RNN

$$\begin{pmatrix} r \\ z \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \end{pmatrix} W_1 \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$\tilde{h}_t = \tanh \left(W_2 \begin{pmatrix} r \odot h_{t-1} \\ x_t \end{pmatrix} \right)$$

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t$$

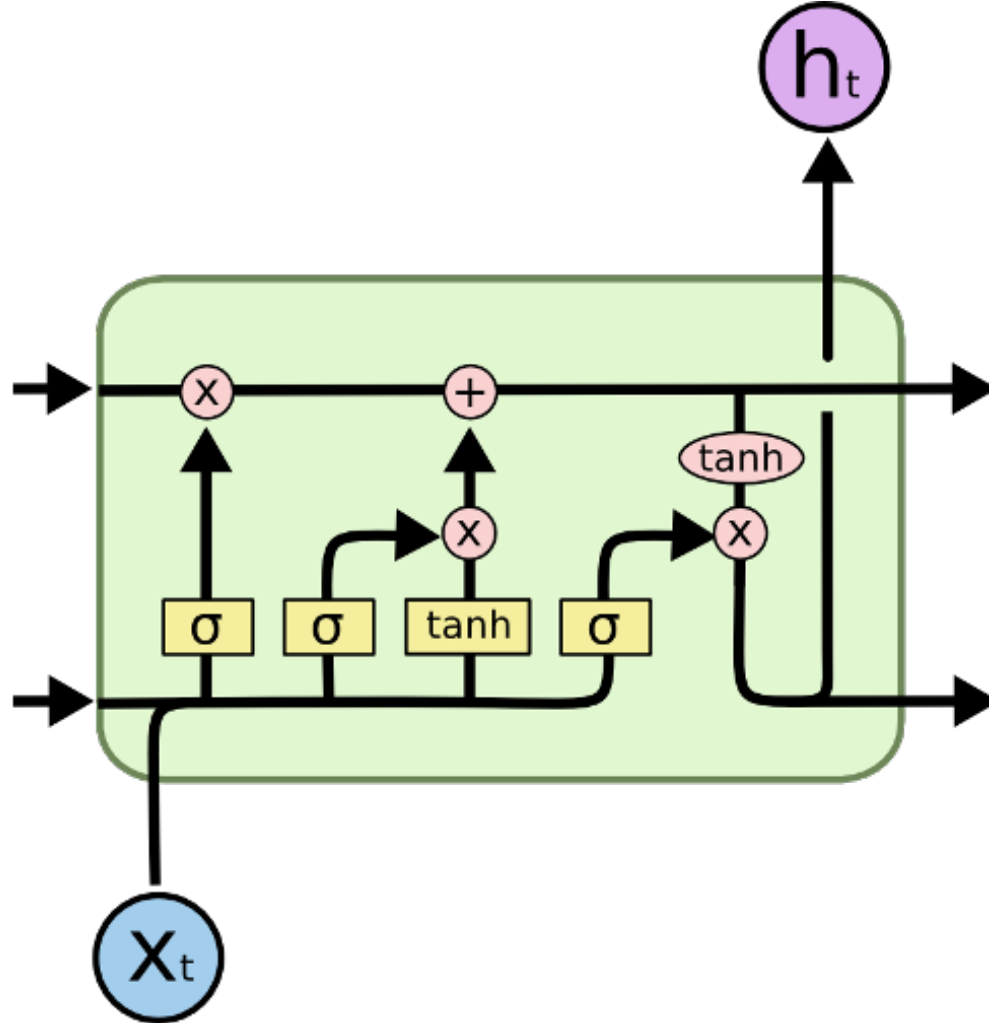


GRU

$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

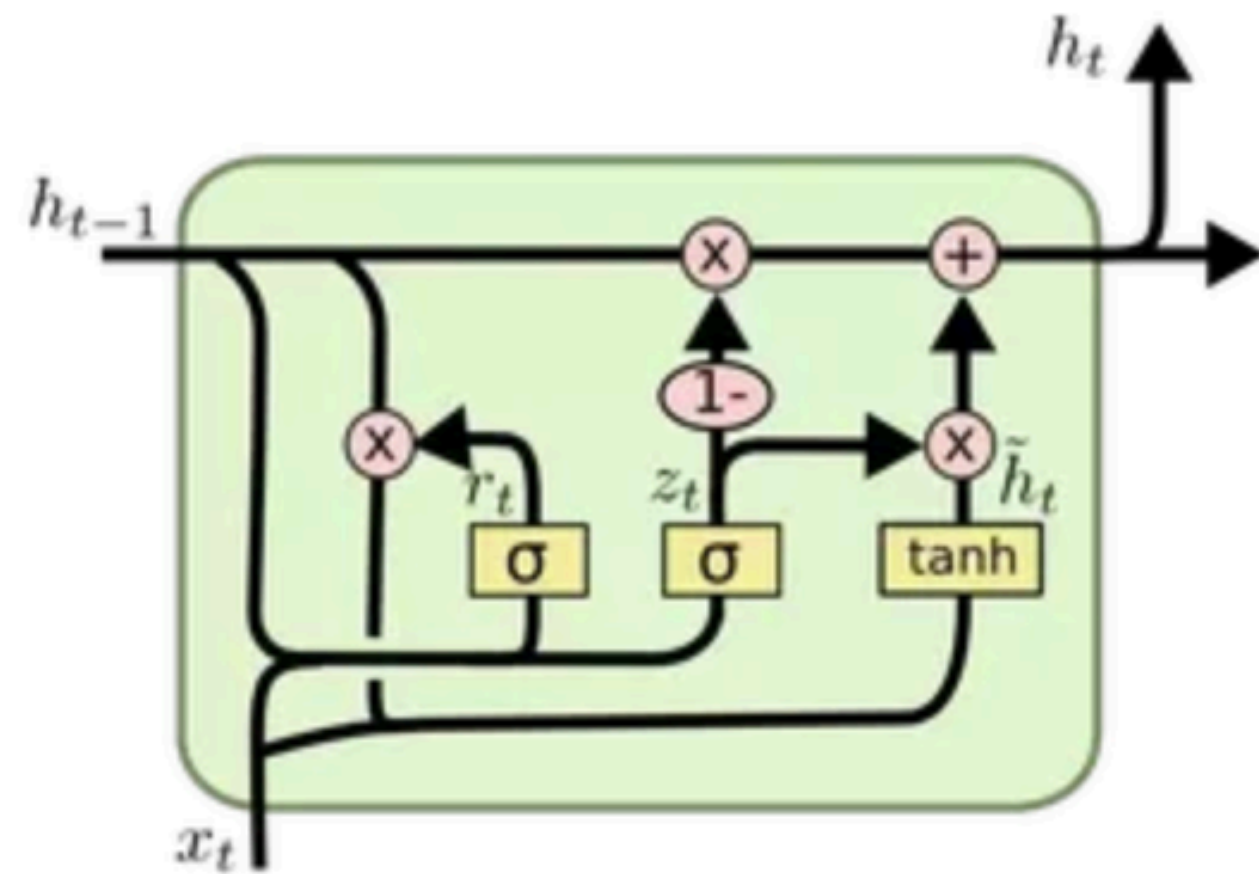
$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$



LSTM

GRU



$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

update

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

reset

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

final hidden state

- If reset is close to 0, ignore previous hidden state
 - Allows model to drop information that is irrelevant in the future
- Update gate z controls how much of past state should matter now.
 - If z close to 1, then we can copy information in that unit through many time steps! Less vanishing gradient!
- Units with short-term dependencies often have reset gates very active

Progress on language models

On the Penn Treebank (PTB) dataset

Metric: [perplexity](#)

KN5: Kneser-Ney 5-gram

Model	Individual
KN5	141.2
KN5 + cache	125.7
Feedforward NNLM	140.2
Log-bilinear NNLM	144.5
Syntactical NNLM	131.3
Recurrent NNLM	124.7
RNN-LDA LM	113.7

$$\text{ppl}(S) = 2^x \quad \text{where}$$
$$x = -\frac{1}{W} \sum_{i=1}^n \log_2 P(S^i)$$

Progress on language models

On the Penn Treebank (PTB) dataset

Metric: [perplexity](#)

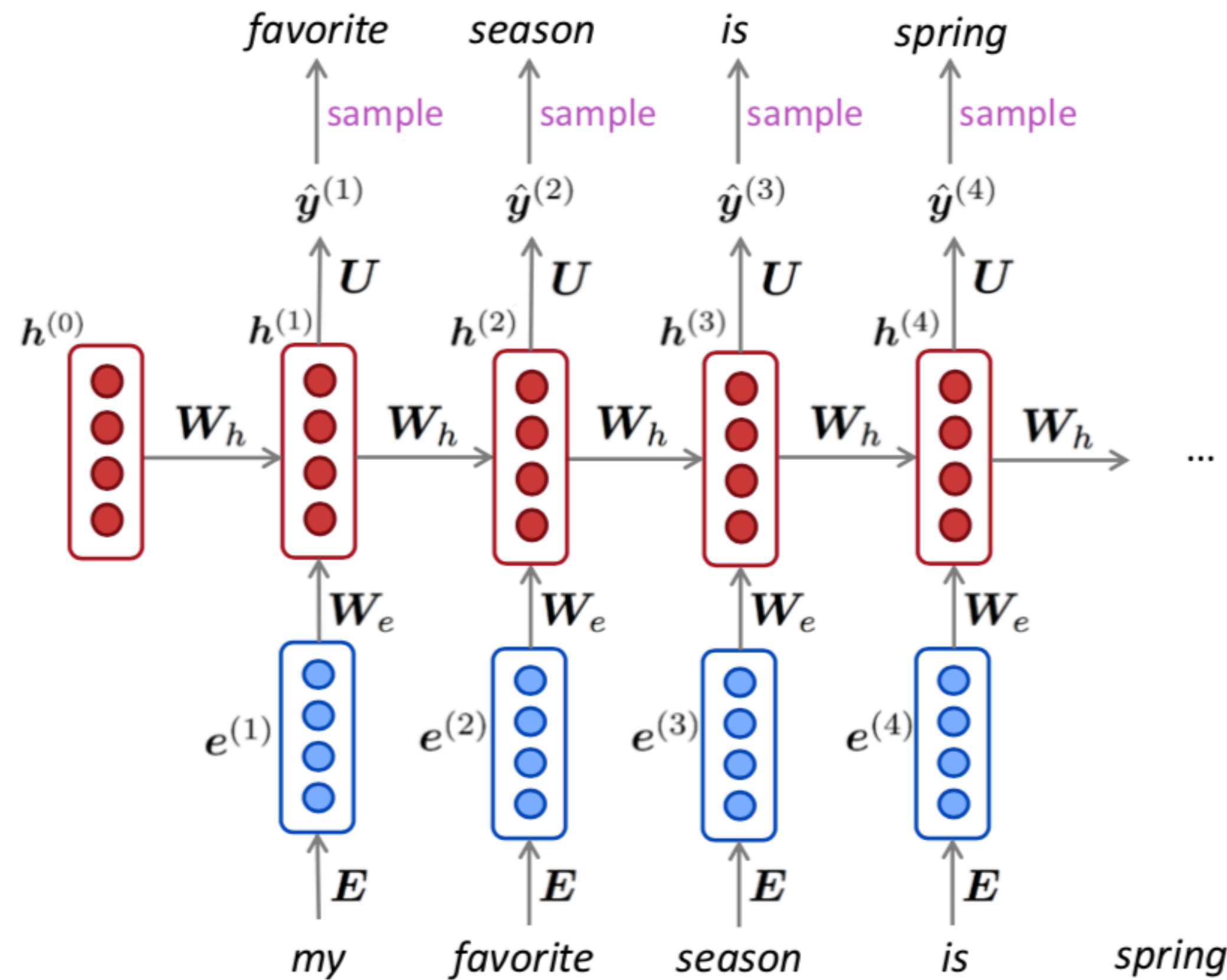
Model	#Param	Validation	Test
Mikolov & Zweig (2012) – RNN-LDA + KN-5 + cache	9M [‡]	-	92.0
Zaremba et al. (2014) – LSTM	20M	86.2	82.7
Gal & Ghahramani (2016) – Variational LSTM (MC)	20M	-	78.6
Kim et al. (2016) – CharCNN	19M	-	78.9
Merity et al. (2016) – Pointer Sentinel-LSTM	21M	72.4	70.9
Grave et al. (2016) – LSTM + continuous cache pointer [†]	-	-	72.1
Inan et al. (2016) – Tied Variational LSTM + augmented loss	24M	75.7	73.2
Zilly et al. (2016) – Variational RHN	23M	67.9	65.4
Zoph & Le (2016) – NAS Cell	25M	-	64.0
Melis et al. (2017) – 2-layer skip connection LSTM	24M	60.9	58.3
Merity et al. (2017) – AWD-LSTM w/o finetune	24M	60.7	58.8
Merity et al. (2017) – AWD-LSTM	24M	60.0	57.3
Ours – AWD-LSTM-MoS w/o finetune	22M	58.08	55.97
Ours – AWD-LSTM-MoS	22M	56.54	54.44
Merity et al. (2017) – AWD-LSTM + continuous cache pointer [†]	24M	53.9	52.8
Krause et al. (2017) – AWD-LSTM + dynamic evaluation [†]	24M	51.6	51.1
Ours – AWD-LSTM-MoS + dynamic evaluation [†]	22M	48.33	47.69

(Yang et al, 2018): Breaking the Softmax Bottleneck: A High-Rank RNN Language Model

Overview

- Review of Vanilla RNN
- Issues with Gradient Flows
- LSTMs and GRUs
- Applications
- Variants: Stacked RNNs, Bidirectional RNNs

Application: Text Generation



You can generate text by **repeated sampling**.
Sampled output is next step's input.

Fun with RNNs

Obama speeches

Good afternoon. God bless you.

The United States will step up to the cost of a new challenges of the American people that will share the fact that we created the problem. They were attacked and so that they have to say that all the task of the final days of war that I will not be able to get this done. The promise of the men and women who were still going to take out the fact that the American people have fought to make sure that they have to be able to protect our part. It was a chance to stand together to completely look for the commitment to borrow from the American people. And the fact is the men and women in uniform and the millions of our country with the law system that we should be a strong stretchs of the forces that we can afford to increase our spirit of the American people and the leadership of our country who are on the Internet of American lives.

Thank you very much. God bless you, and God bless the United States of America.

Latex generation

```
\begin{proof}
We may assume that  $\mathcal{I}$  is an abelian sheaf on  $\mathcal{C}$ .
\item Given a morphism  $\Delta : \mathcal{F} \rightarrow \mathcal{I}$ 
is an injective and let  $\mathfrak{q}$  be an abelian sheaf on  $X$ .
Let  $\mathcal{F}$  be a fibered complex. Let  $\mathcal{F}$  be a category.
\begin{enumerate}
\item \hyperref[setain-construction-phantom]{Lemma}
\label{lemma-characterize-quasi-finite}
Let  $\mathcal{F}$  be an abelian quasi-coherent sheaf on  $\mathcal{C}$ .
Let  $\mathcal{F}$  be a coherent  $\mathcal{O}_X$ -module. Then
 $\mathcal{F}$  is an abelian catenary over  $\mathcal{C}$ .
\item The following are equivalent
\begin{enumerate}
\item  $\mathcal{F}$  is an  $\mathcal{O}_X$ -module.
\end{enumerate}
\end{lemma}
```

